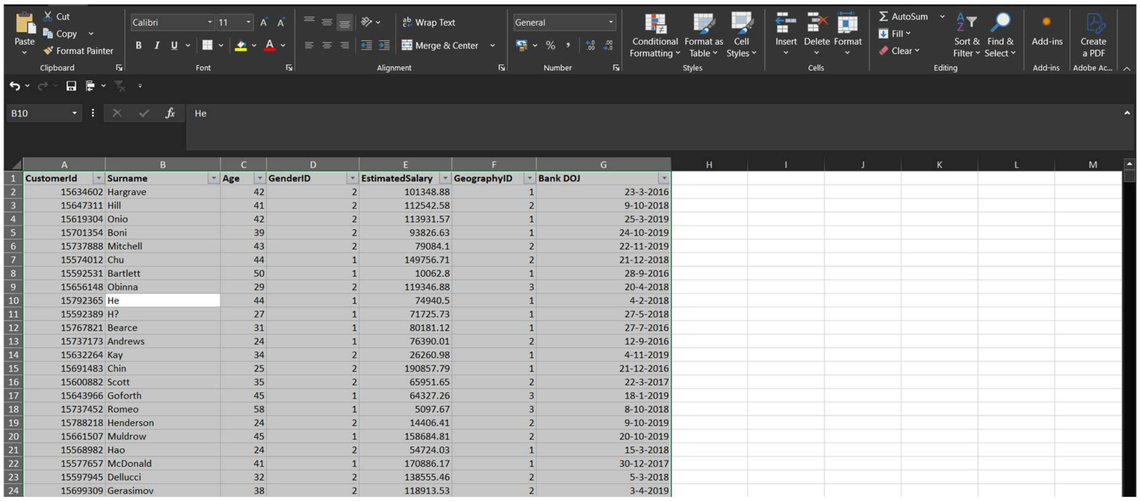
OVERVIEW

- Data Cleaning and Formatting in Excel
- Data Loading and Modelling in SQL
- Data Loading and Modelling in Power BI
- Objective Question
- Subjective Question

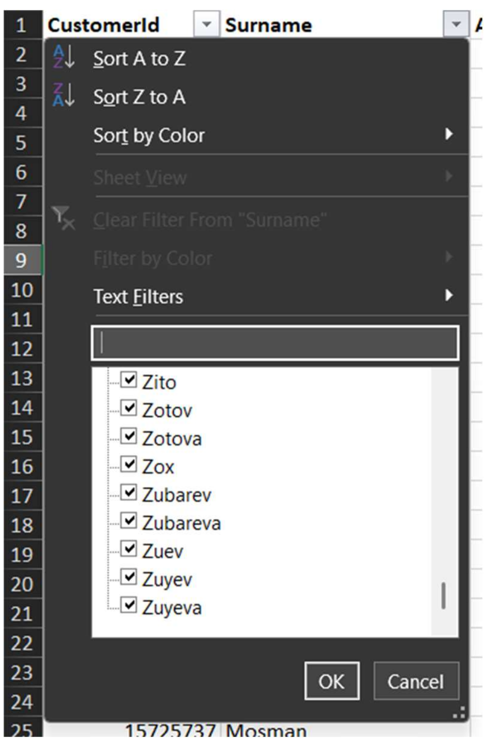
EXCEL: DATA CLEANING AND FORMATTING

1. Excel files: ActiveCustomer, CreditCard, Gender, Geography, and ExitCustomer are already clean, consistent, formatted, and free of duplicates.
2. Cleaning and formatting data in the CustomerInfo file

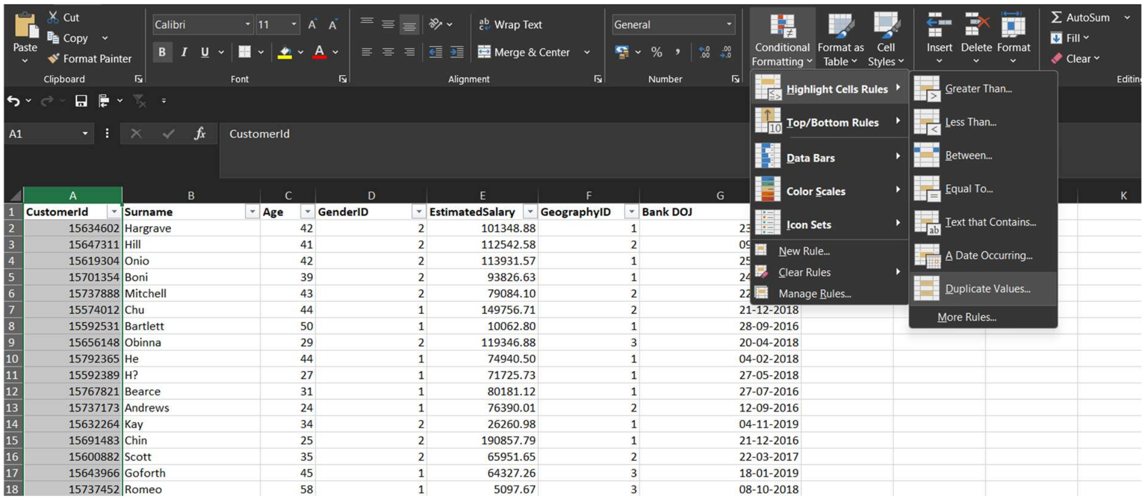
- ➢ First, select all columns in the data, and then enable the filter feature



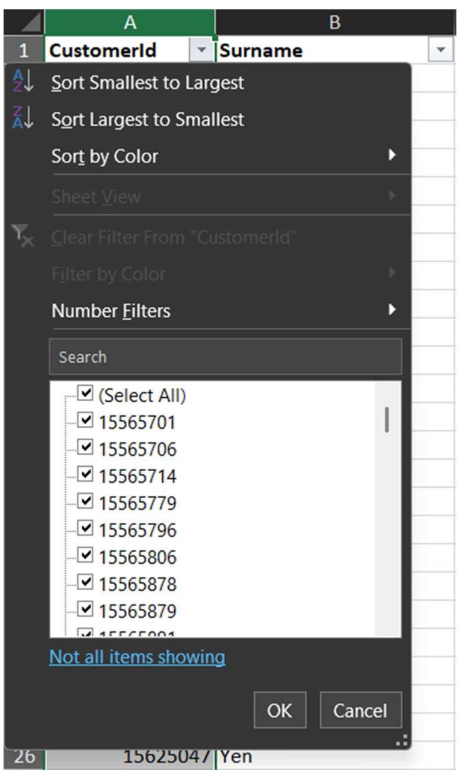
- ➢ Click the dropdown of every column and check if there are any blanks in the column (all the columns have no blanks in any rows)



- ➢ Now using column CustomerId, which is the Primary Key for this table, to find duplicates by selecting the whole column and using conditional formatting to highlight cells based on duplicates



- ➢ Then, in the filter option of the same column, we can see the filter by color option is not enabled, indicating that no duplicates were there (since we have coloured all the duplicates if they exist)



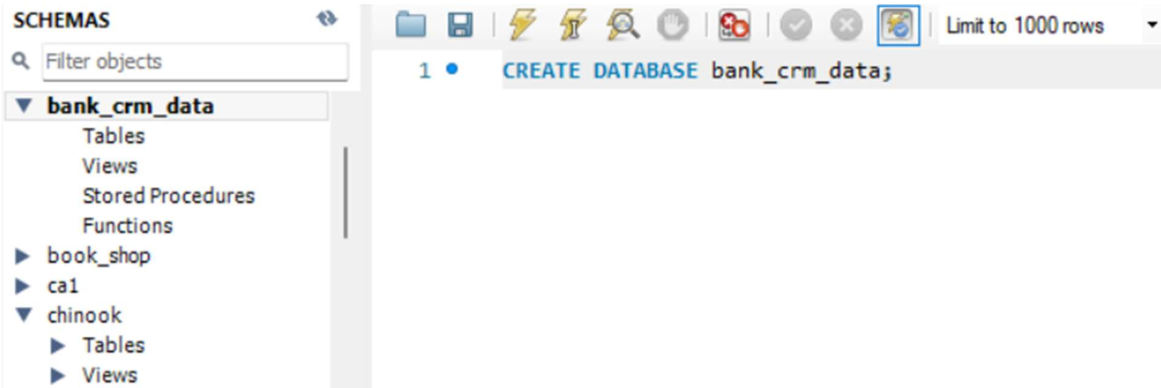
- ➢ Finally changing the datatype of the column Bank DOJ to short date and Estimated Salary to Number with 2 decimals, and so on
- ➢ Now all the data in this file is clean and consistent to be used for further analysis

3. Doing the steps given in point 2 for Bank_Churn table, making it clean, consistent, and formatted
4. After making all tables clean, consistent, and in proper format, converting the previous Excel files into CSV (using the Save As feature) and renaming the table as per the convenience in loading for both Excel and SQL, as follows:

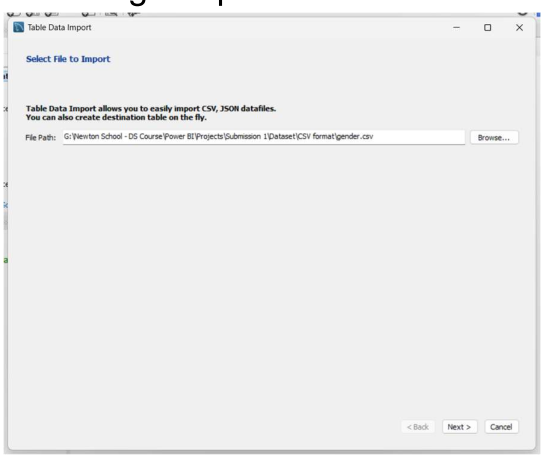
Previous Table Name	**Updated Table Name**
ActiveCustomer	customer_category
Bank_Churn	bank_churn
CreditCard	credit_category
CustomerInfo	customer_detail
ExitCustomer	churn_detail
Gender	gender
Geography	geography

SQL: DATA LOADING AND MODELLING

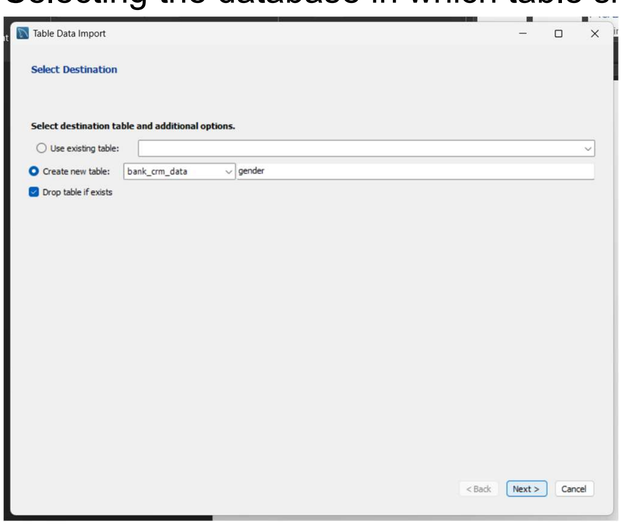
1. Creating a database “bank_crm_data” in SQL using the following query



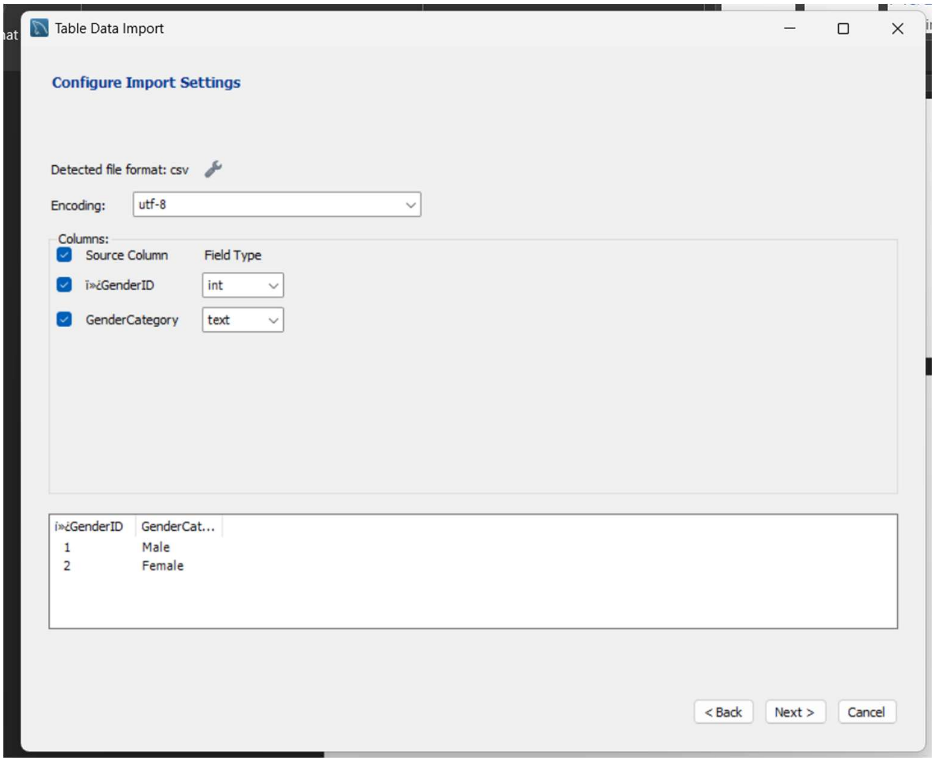
2. **Data Loading Process:**
   Right-clicking the database and then using the Table Data Import Wizard
   a. Selecting the path of CSV files we earlier made



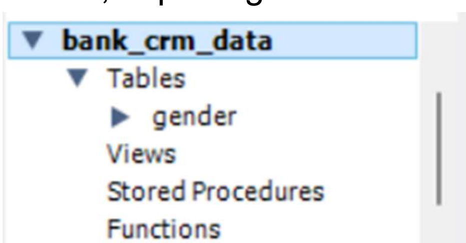
   b. Selecting the database in which table should be placed



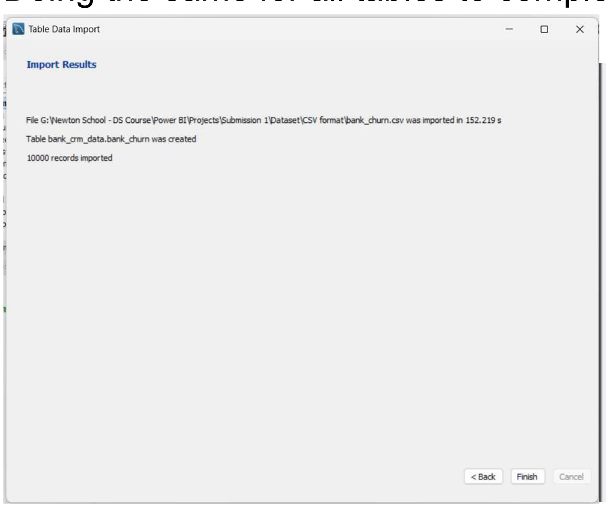
c. Selecting data formatting of table in right format

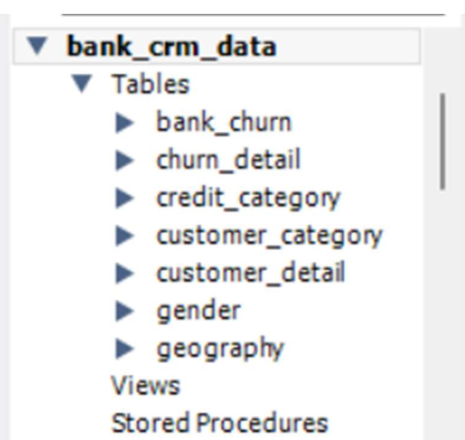


d. At last, importing the data into the database



e. Doing the same for all tables to complete the dataset





3. Data Cleaning and Formatting

a. In the customer_detail table, the column “Bank DOJ” was imported as text, so we will first rename the column to remove the space (making querying easier) and then change the data type to datetime using the STR_TO_DATE function.

```
-- manipulating the column in customer_detail table
ALTER TABLE customer_detail
RENAME COLUMN `Bank DOJ` TO BankDOJ;

UPDATE customer_detail
SET BankDOJ = STR_TO_DATE(BankDOJ, '%d-%m-%Y');

ALTER TABLE customer_detail
MODIFY COLUMN BankDOJ DATE;

DESC customer_detail;

SELECT *
FROM customer_detail;
```

output

ï»¿CustomerId	Surname	Age	GenderID	EstimatedSalary	GeographyID	BankDOJ
15634602	Hargrave	42	2	101348.88	1	2016-03-23
15647311	Hill	41	2	112542.58	2	2018-10-09
15619304	Onio	42	2	113931.57	1	2019-03-25
15701354	Boni	39	2	93826.63	1	2019-10-24
15737888	Mitchell	43	2	79084.1	2	2019-11-22
15574012	Chu	44	1	149756.71	2	2018-12-21
15592531	Bartlett	50	1	10062.8	1	2016-09-28
15656148	Obinna	29	2	119346.88	3	2018-04-20
15792365	He	44	1	74940.5	1	2018-02-04
15592389	H?	27	1	71725.73	1	2018-05-27

description

Field	Type	Null	Key	Default	Extra
ï»¿CustomerId	int	YES		NULL	
Surname	text	YES		NULL	
Age	int	YES		NULL	
GenderID	int	YES		NULL	
EstimatedSalary	double	YES		NULL	
GeographyID	int	YES		NULL	
BankDOJ	date	YES		NULL	

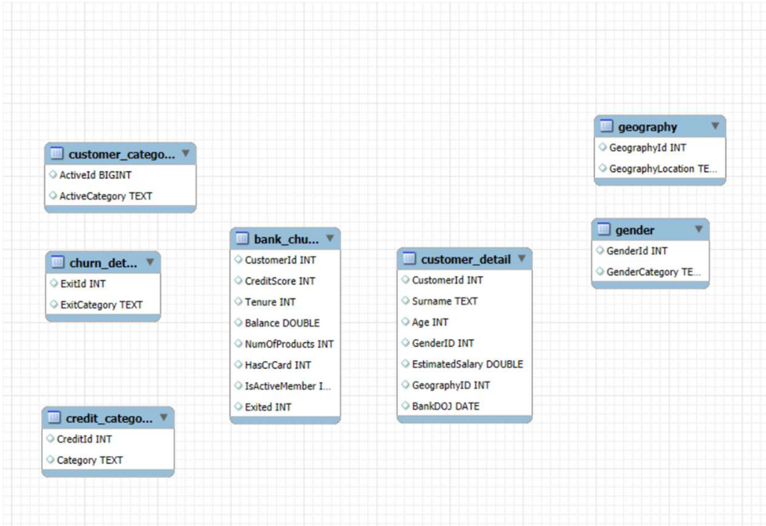
b. As we can see, all tables have their first column name starting from unwanted character “ï»¿” so we will rename all column names

```sql
-- renaming first column of every table in right format
ALTER TABLE bank_churn
RENAME COLUMN ï»¿CustomerId TO CustomerId;

ALTER TABLE churn_detail
RENAME COLUMN ï»¿ExitID TO ExitId;

ALTER TABLE credit_category
RENAME COLUMN ï»¿CreditID TO CreditId;

ALTER TABLE customer_category
RENAME COLUMN ï»¿ActiveID TO ActiveId;

ALTER TABLE customer_detail
RENAME COLUMN ï»¿CustomerId TO CustomerId;

ALTER TABLE gender
RENAME COLUMN ï»¿GenderID TO GenderId;

ALTER TABLE geography
RENAME COLUMN ï»¿GeographyID TO GeographyId;
```

sample output of one table

	CustomerId	Surname	Age	GenderID	EstimatedSalary	GeographyID	BankDOJ

4. **Data Relationship Establishment:**
   a. Here is the ERD of the current database

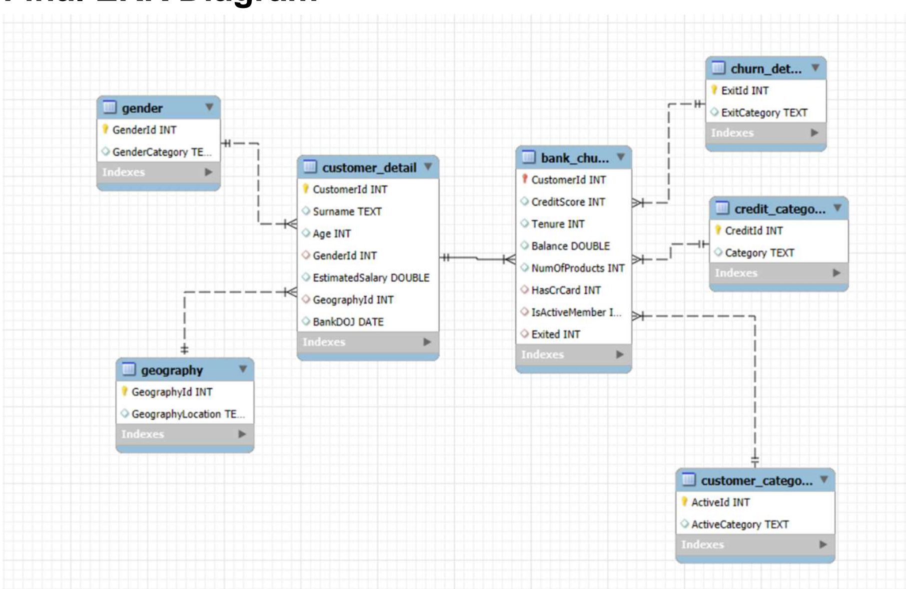


   b. First, we will assign a primary key column to every table, and then we will use these columns to develop relationship to other columns using a foreign key

```sql
-- making primary key using alter
ALTER TABLE bank_churn
ADD CONSTRAINT PRIMARY KEY (CustomerId);

ALTER TABLE churn_detail
ADD CONSTRAINT PRIMARY KEY (ExitId);

ALTER TABLE credit_category
ADD CONSTRAINT PRIMARY KEY (CreditId);

ALTER TABLE customer_category
ADD CONSTRAINT PRIMARY KEY (ActiveId);

ALTER TABLE customer_detail
ADD CONSTRAINT PRIMARY KEY (CustomerId);

ALTER TABLE gender
ADD CONSTRAINT PRIMARY KEY (GenderId);

ALTER TABLE geography
ADD CONSTRAINT PRIMARY KEY (GeographyId);
```

```sql
-- establishing relatiohship amongst tables
ALTER TABLE bank_churn
ADD CONSTRAINT FK_to_ChurnDetail
FOREIGN KEY (Exited)
REFERENCES churn_detail(ExitId);

ALTER TABLE bank_churn
ADD CONSTRAINT FK_to_CreditCategory
FOREIGN KEY (HasCrCard)
REFERENCES credit_category(CreditId);

ALTER TABLE customer_category
MODIFY COLUMN ActiveId INT;   -- previously it was BIGINT

ALTER TABLE bank_churn
ADD CONSTRAINT FK_to_CustomerCategory
FOREIGN KEY (IsActiveMember)
REFERENCES customer_category(ActiveId);

ALTER TABLE customer_detail
ADD CONSTRAINT FK_to_Gender
FOREIGN KEY (GenderId)
REFERENCES Gender(GenderId);

ALTER TABLE customer_detail
ADD CONSTRAINT FK_to_Geography
FOREIGN KEY (GeographyId)
REFERENCES geography(GeographyId);

ALTER TABLE bank_churn
ADD CONSTRAINT FK_to_CustomerDetail
FOREIGN KEY (CustomerId)
REFERENCES customer_detail(CustomerId);
```

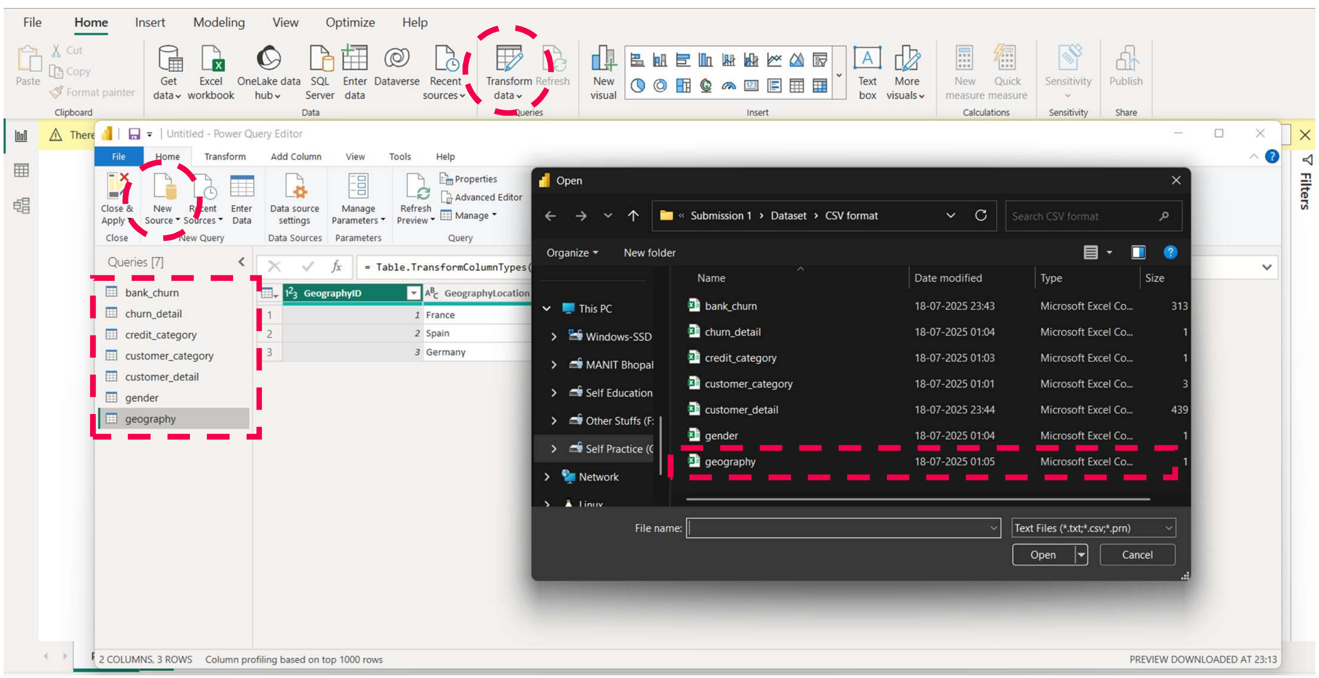
Final ERR Diagram



POWER BI: DATA LOADING AND MODELLING

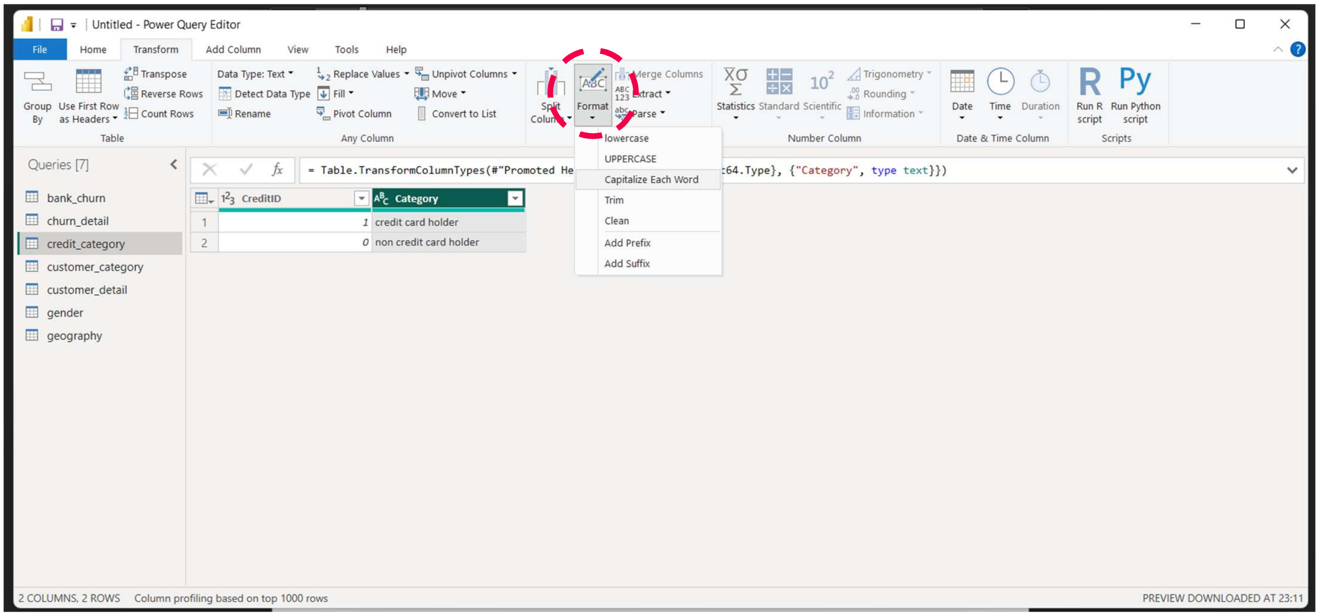
1. Data Loading

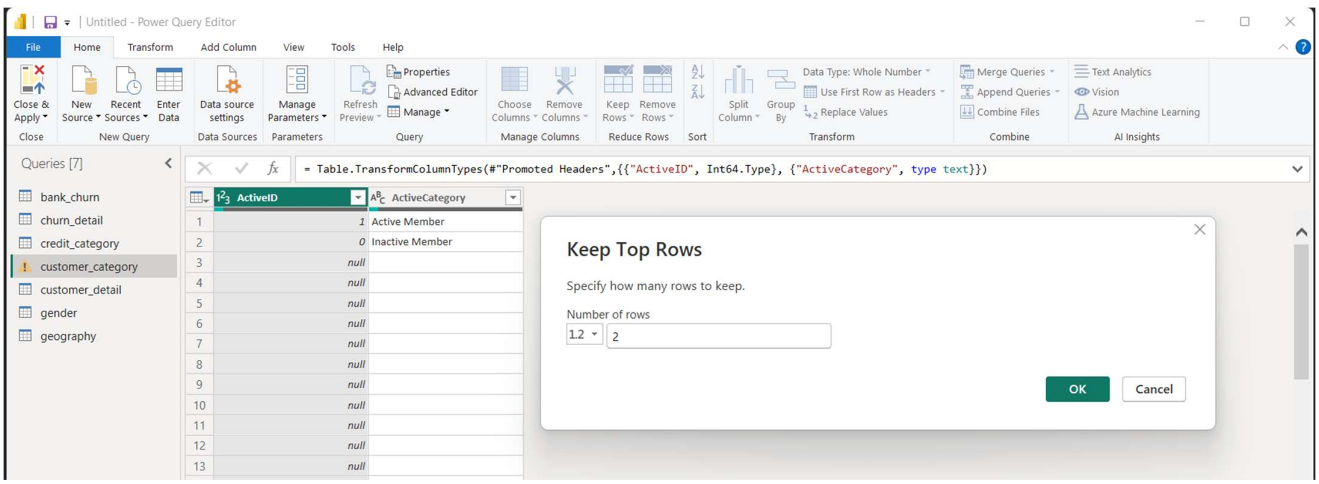
- Open a new Power BI file
- Select “Transform Data” in the Home tab to open Power Query Editor
- Then click on “New Sources” and select CSV, choosing the destination of the file to be used as the source (in our case, the same files we used for the SQL section)
- Using these steps we have loaded in all our required tables in the file

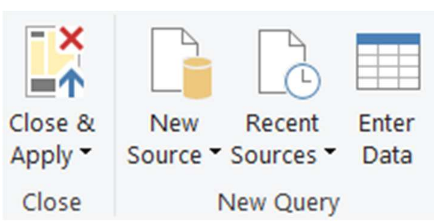


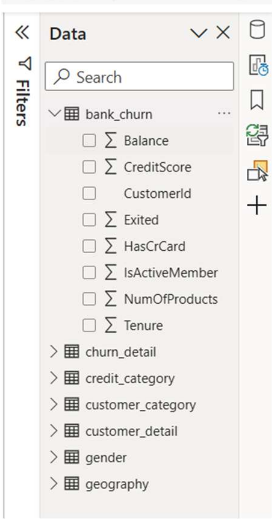
2. Data Cleaning and Manipulation

- Tables bank_churn, churn_detail, customer_detail, gender, and geography are all cleaned and consistent, with the correct data types assigned to them
- In the table credit_category we will manipulate the column Category to make each word capitalized by using the “Transform” tab’s “Format” Tool
- In the table customer_category there exist some unwanted rows as null values so using “Keep Top Rows” tool we will only intake first two rows of data that is useful to us
- At last using “Close & Apply” to load all the tables in Power BI


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Group By
Use First Row as Headers
Transpose
Reverse Rows
Count Rows
Table
Data Type: Text
Detect Data Type
Rename
Replace Values
Fill
Pivot Column
Unpivot Columns
Move
Convert to List
Any Column
Split Column
Format
Merge Columns
Extract
Parse
lowercase
UPPERCASE
Capitalize Each Word
Trim
Clean
Add Prefix
Add Suffix
Statistics
Standard
Scientific
Trigonometry
Rounding
Information
Number Column
Date
Time
Duration
Date & Time Column
Run R script
Run Python script
Scripts
Queries [7]
bank_churn
churn_detail
credit_category
customer_category
customer_detail
gender
geography
= Table.TransformColumnTypes(#"Promoted He
:64.Type}, {"Category", type text}})
CreditID
Category
1
credit card holder
0
non credit card holder
2 COLUMNS, 2 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 23:11

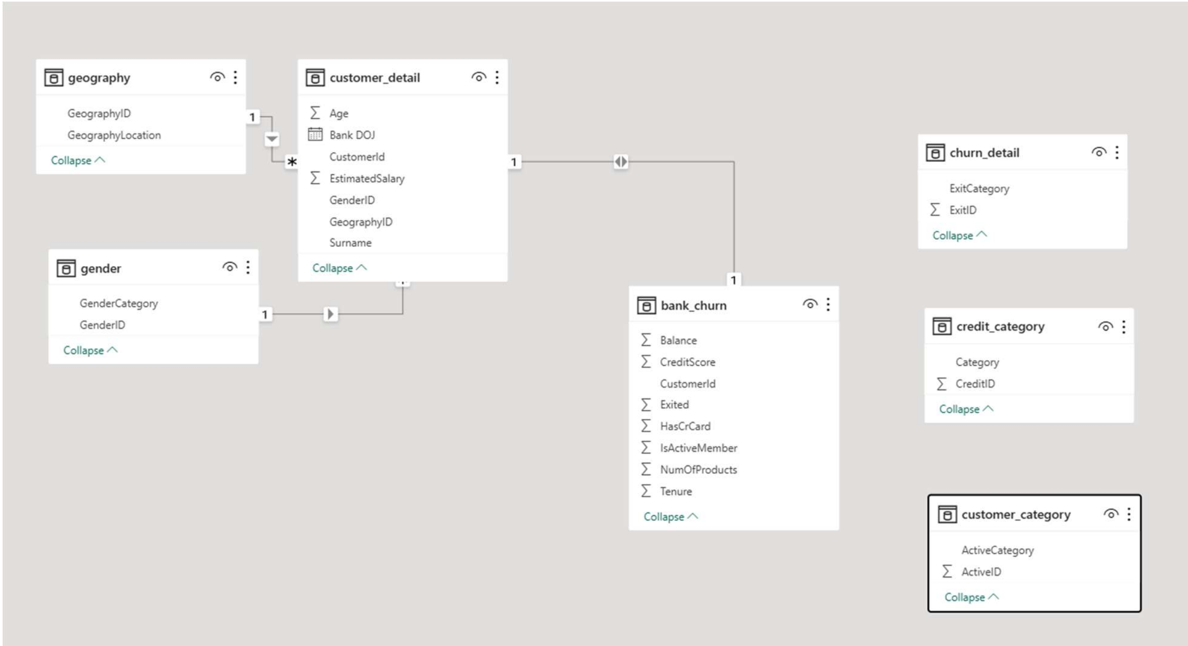

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Whole Number
Use First Row as Headers
Replace Values
Merge Queries
Append Queries
Combine Files
Text Analytics
Vision
Azure Machine Learning
Close
New Query
Data Sources
Parameters
Query
Manage Columns
Reduce Rows
Sort
Transform
Combine
AI Insights
Queries [7]
bank_churn
churn_detail
credit_category
customer_category
customer_detail
gender
geography
= Table.TransformColumnTypes(#"Promoted Headers",{{"ActiveID", Int64.Type}, {"ActiveCategory", type text}})
ActiveID
ActiveCategory
Active Member
Inactive Member
null
Keep Top Rows
Specify how many rows to keep.
Number of rows
2
OK
Cancel


Close & Apply
New Source
Recent Sources
Enter Data
Close
New Query

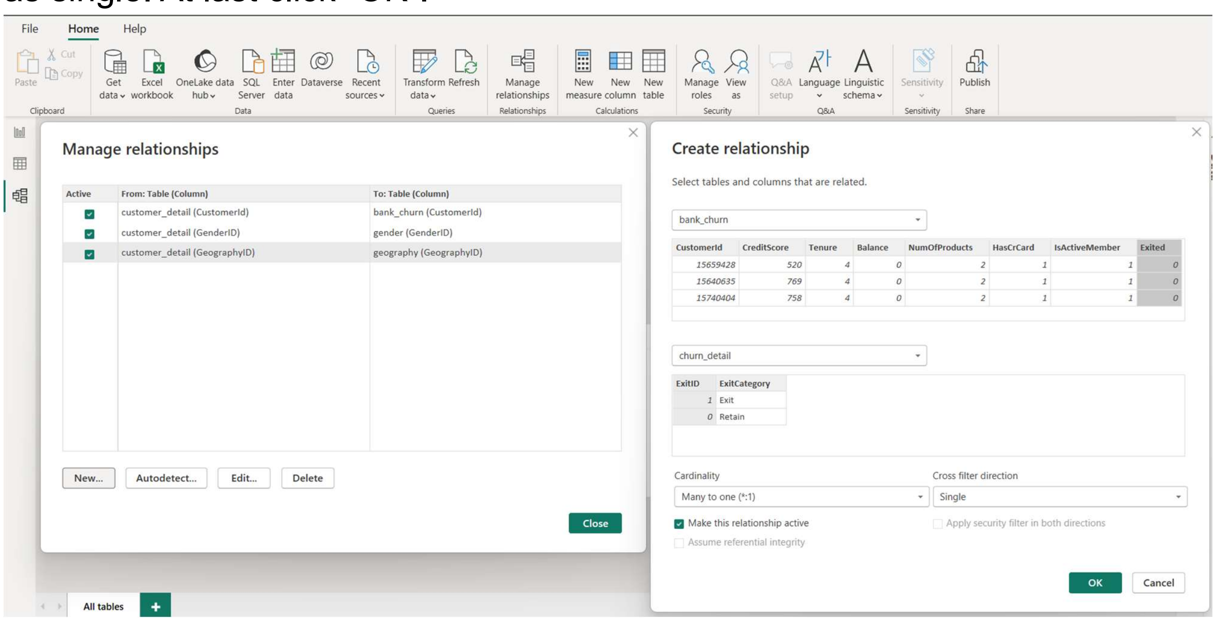

Data
Filters
Search
bank_churn
Balance
CreditScore
CustomerId
Exited
HasCrCard
IsActiveMember
NumOfProducts
Tenure
churn_detail
credit_category
customer_category
customer_detail
gender
geography

3. Data Cleaning and Manipulation

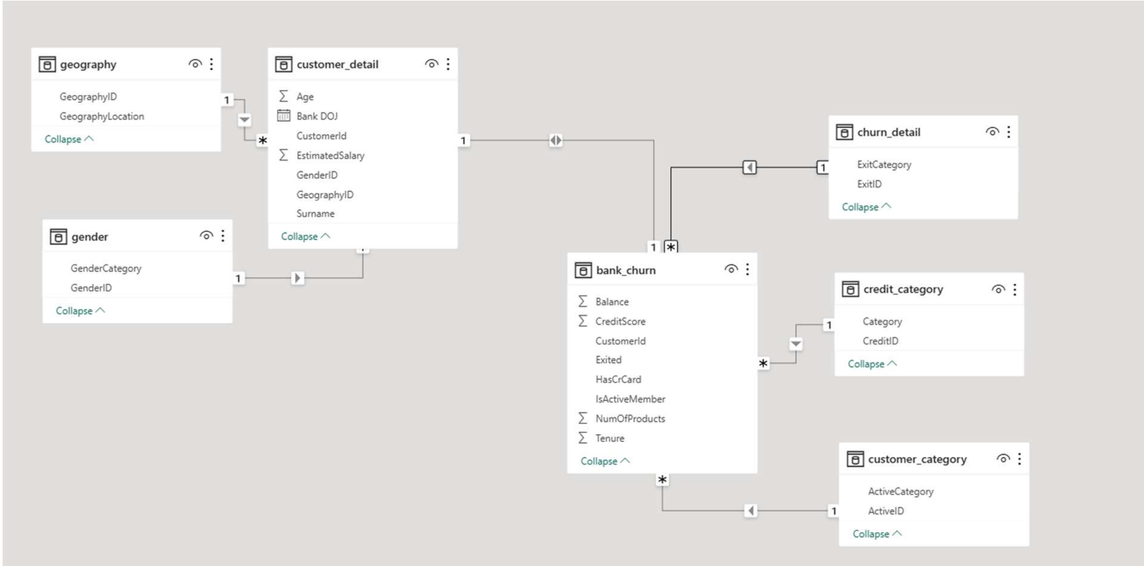
- In data model view we can see the current state of our database to find relationships amongst tables. Power BI detected some of the relationships but couldn't develop other because of different column names



- Now we will use the "Manage Relationship" tool to add a "New" relationship. Select the first and the second table. Click columns that will be used to establish relationship. Then verify cardinality and keep cross-filter direction as single. At last click "OK".



- After establishing all relationships our data model looks as follows



OBJECTIVE QUESTIONS

```sql
CREATE VIEW bank_cust_details AS (
  SELECT
    c.CustomerId,
        c.Surname,
        c.Age,
        g.GenderCategory AS Gender,
        gp.GeographyLocation AS Location,
        c.BankDOJ,
        c.EstimatedSalary,
        b.Balance,
        b.Tenure,
        b.CreditScore,
        b.IsActiveMember,
        b.HasCrCard,
        b.NumOfProducts,
        b.Exited
    FROM bank_churn b
    JOIN customer_detail c
    ON b.CustomerId = c.CustomerId
  JOIN gender g
    ON c.GenderId = g.GenderId
  JOIN geography gp
    ON gp.GeographyId = c.GeographyId
);

SELECT *
FROM bank_cust_details;
```

CustomerId	Surname	Age	Gender	Location	BankDOJ	EstimatedSalary	Balance	Tenure	CreditScore	IsActiveMember	HasCrCard	NumOfProducts	Exited
15565879	Riley	28	Female	France	2017-11-23	56185.98	0	5	845	1	1	2	0
15566139	Ts'ui	37	Female	France	2016-09-14	62830.97	53573.18	7	526	0	1	1	0
15566251	Ferrari	37	Female	France	2018-10-01	98686.4	96652.86	4	618	0	1	1	1
15566295	Sanders	33	Female	France	2018-04-29	148779.41	138053.79	5	761	0	1	2	0
15566563	Duigan	30	Female	France	2017-09-29	5008.23	137851.31	5	777	0	1	1	1

1. **What is the distribution of account balances across different regions?**

```sql
SELECT
    percentile,
    SUM(CASE WHEN GeographyLocation = "France" THEN EstimatedSalary END) AS FranceRegion,
    SUM(CASE WHEN GeographyLocation = "Spain" THEN EstimatedSalary END) AS SpainRegion,
    SUM(CASE WHEN GeographyLocation = "Germany" THEN EstimatedSalary END) AS GermanyRegion
FROM (
    SELECT
        g.GeographyId,
        g.GeographyLocation,
        EstimatedSalary,
        ROUND(perrnk, 2) AS percentile
    FROM (
        SELECT *,
            ROUND(PERCENT_RANK() OVER(PARTITION BY GeographyId ORDER BY EstimatedSalary), 4) AS perrnk
        FROM customer_detail
    ) r
    JOIN geography g
        ON r.GeographyId = g.GeographyId
    WHERE perrnk IN (0, 0.25, 0.5, 0.5001, 0.75, 1)
) prnk
GROUP BY percentile;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

percentile	FranceRegion	SpainRegion	GermanyRegion
0	90.07	417.41	11.58
0.25	51380.9	50267.69	51016.02
0.5	99136.49	99984.86	102397.22
0.75	149331.01	147278.43	151083.8
1	199929.17	199992.48	199970.74

2. **Identify the top 5 customers with the highest Estimated Salary in the last quarter of the year. (SQL)**

```sql
SELECT
    sr.CustomerId,
    sr.Surname,
    sr.Age,
    g.GenderCategory,
    y.GeographyLocation,
    sr.BankDOJ,
    sr.EstimatedSalary
FROM (
    SELECT *,
        DENSE_RANK() OVER(ORDER BY EstimatedSalary DESC) AS rnk
    FROM customer_detail
    WHERE EXTRACT(YEAR FROM BankDOJ) = (SELECT DISTINCT EXTRACT(YEAR FROM MAX(BankDOJ)) FROM customer_detail)
        AND EXTRACT(QUARTER FROM BankDOJ) = (SELECT DISTINCT EXTRACT(QUARTER FROM MAX(BankDOJ)) FROM customer_detail)
) sr
JOIN gender g
    ON sr.GenderId = g.GenderId
JOIN geography y
    ON sr.GeographyId = y.GeographyId
WHERE rnk <= 5
ORDER BY EstimatedSalary DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerId	Surname	Age	GenderCategory	GeographyLocation	BankDOJ	EstimatedSalary
15804211	Oluchukwu	36	Male	France	2019-12-25	199841.32
15763065	Palerma	40	Female	Spain	2019-10-01	199753.97
15624641	Kharlamova	43	Male	Spain	2019-10-02	199290.68
15691874	Kazakova	34	Female	France	2019-12-19	198929.84
15699095	Chandler	24	Female	France	2019-12-15	198826.03

3. **Calculate the average number of products used by customers who have a credit card. (SQL)**

```sql
SELECT ROUND(AVG(NumOfProducts)) AS AvgNumOfProducts
FROM bank_churn
WHERE HasCrCard = 1;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

AvgNumOfProducts
2

4. **Determine the churn rate by gender for the most recent year in the dataset.**

```sql
WITH base_table AS (
    SELECT *
    FROM bank_cust_details
    WHERE EXTRACT(YEAR FROM BankDOJ) = (SELECT DISTINCT EXTRACT(YEAR FROM MAX(BankDOJ)) FROM bank_cust_details)
)

SELECT
    Gender,
    ROUND((COUNT(CASE WHEN Exited=1 THEN CustomerId END)*100 / COUNT(CustomerId)), 2) AS recent_churn_rate
FROM base_table
```

Result Grid Filter Rows: Export: Wrap Cell Content:

Gender	recent_churn_rate
Female	25.05
Male	15.37

5. **Compare the average credit score of customers who have exited and those who remain. (SQL)**

Average Credit Score of Retained customer is higher than Exited Customer

```sql
SELECT
    c.ExitCategory,
    ROUND(AVG(b.CreditScore)) AS AvgCreditScore
FROM bank_cust_details b
JOIN churn_detail c
    ON c.ExitId = b.Exited
GROUP BY c.ExitCategory;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

ExitCategory	AvgCreditScore
Retain	652
Exit	645

6. **Which gender has a higher average estimated salary, and how does it relate to the number of active accounts? (SQL)**

Females have a **higher** average estimated salary and a **higher** number of active accounts

```sql
SELECT
    Gender,
    ROUND(AVG(EstimatedSalary), 2) AS AvgEstimatedSalary,
    COUNT(CASE WHEN IsActiveMember=0 THEN CustomerId END) AS CntOfActiveAccount,
    ROUND((COUNT(CASE WHEN IsActiveMember=0 THEN CustomerId END)*100 / COUNT(CustomerId)), 2) AS PrctOfActiveAccount
FROM bank_cust_details
GROUP BY Gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Gender	AvgEstimatedSalary	CntOfActiveAccount	PrctOfActiveAccount
Female	100601.54	2259	49.72
Male	99664.58	2590	47.46

7. **Segment the customers based on their credit score and identify the segment with the highest exit rate. (SQL)**

Fair Credit Customer Segment has the highest overall Exited Customer, about 6.85% Poor Credit Customer Segment has the highest exited customers, about 22.02% segment-wise

```sql
SELECT
    (
        CASE
        WHEN IF(CreditScore<300, 300, CreditScore) BETWEEN 300 AND 579 THEN "300-579 : Poor"
        WHEN CreditScore BETWEEN 580 AND 669 THEN "580-669 : Fair"
        WHEN CreditScore BETWEEN 670 AND 739 THEN "670-739 : Good"
        WHEN CreditScore BETWEEN 740 AND 799 THEN "740-799 : Very Good"
        WHEN IF(CreditScore>850, 850, CreditScore) BETWEEN 800 AND 850 THEN "800-850 : Exceptional"
        END
    ) AS CustomerSegment,
    COUNT(CustomerId) AS CntOfCustomer,
    COUNT(CASE WHEN Exited=1 THEN CustomerId END) AS CntOfExitedCustomer,
    ROUND((COUNT(CASE WHEN Exited=1 THEN CustomerId END)*100 / COUNT(CustomerId)), 2) AS PrctOfExitedCustomer,
    ROUND((COUNT(CASE WHEN Exited=1 THEN CustomerId END)*100 / (SELECT COUNT(CustomerId) FROM bank_cust_details)), 2) AS PrctOutOfTotalCustomer
FROM bank_cust_details
GROUP BY CustomerSegment
ORDER BY CustomerSegment;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerSegment	CntOfCustomer	CntOfExitedCustomer	PrctOfExitedCustomer	PrctOutOfTotalCustomer
300-579 : Poor	2362	520	22.02	5.20
580-669 : Fair	3331	685	20.56	6.85
670-739 : Good	2428	452	18.62	4.52
740-799 : Very Good	1224	252	20.59	2.52
800-850 : Exceptional	655	128	19.54	1.28

8. **Find out which geographic region has the highest number of active customers with a tenure greater than 5 years. (SQL)**

France is the required region

```sql
SELECT
    Location,
    COUNT(CustomerId) AS CntOfCustomer,
    ROUND((COUNT(CustomerId)*100 / (SELECT COUNT(CustomerId) FROM bank_cust_details WHERE IsActiveMember=1 AND Tenure>5)), 2) AS PrcntOfCustomers
FROM bank_cust_details
WHERE IsActiveMember=1 AND Tenure>5
GROUP BY Location
ORDER BY PrcntOfCustomers DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Location	CntOfCustomer	PrcntOfCustomers
France	797	48.99
Spain	431	26.49
Germany	399	24.52

9. **What is the impact of having a credit card on customer churn, based on the available data?**

Having a Credit Card as a factor alone cannot conclude churn of customer because both the customer segments have same amount of Churn

```sql
SELECT
    UPPER((
        SELECT Category FROM credit_category c
        WHERE c.CreditId = b.HasCrCard
    )) AS CustomerSegment,
    COUNT(CASE WHEN Exited=1 THEN CustomerId END) AS CntOfExitedCustomer,
    COUNT(CustomerId) AS CntOfCustomer,
    ROUND((COUNT(CASE WHEN Exited=1 THEN CustomerId END)*100 / COUNT(CustomerId)), 2) AS PrcntOfExitedCustomer
FROM bank_cust_details b
GROUP BY HasCrCard;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerSegment	CntOfExitedCustomer	CntOfCustomer	PrcntOfExitedCustomer
CREDIT CARD HOLDER	1424	7055	20.18
NON CREDIT CARD HOLDER	613	2945	20.81

10. **For customers who have exited, what is the most common number of products they have used?**

About 69.17% of customers who have exited have only 1 product

```sql
SELECT
    NumOfProducts,
    COUNT(CustomerId) AS CntOfCustomer,
    ROUND((COUNT(CustomerId)*100 / (SELECT COUNT(CustomerId) FROM bank_cust_details WHERE Exited=1)), 2) AS PrcntOfCustomers
FROM bank_cust_details
WHERE Exited = 1
GROUP BY NumOfProducts
ORDER BY PrcntOfCustomers DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

NumOfProducts	CntOfCustomer	PrcntOfCustomers
1	1409	69.17
2	348	17.08
3	220	10.80
4	60	2.95

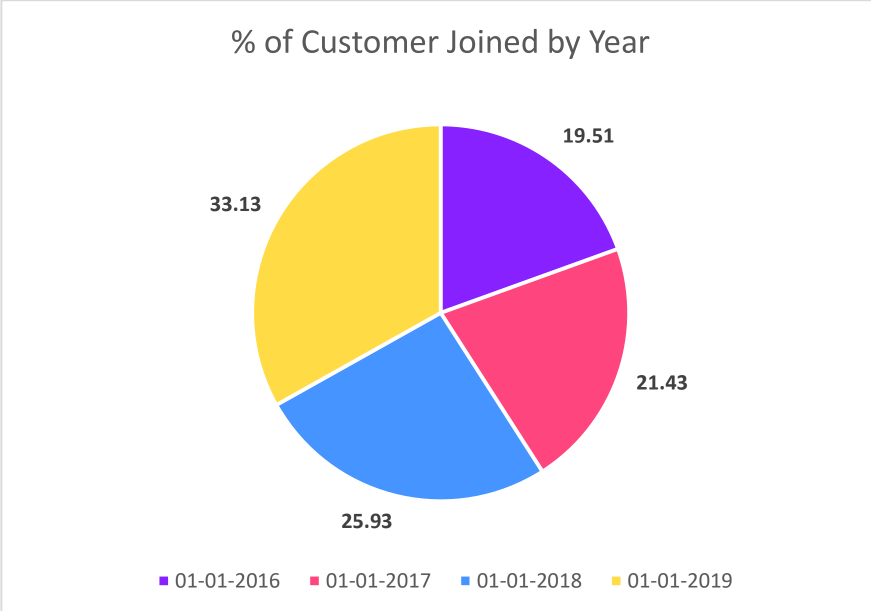
11. **Examine the trend of customers joining over time and identify any seasonal patterns (yearly or monthly). Prepare the data through SQL and then visualize it.**

YEARLY

```sql
-- Yearly
SELECT
    Years AS JoiningYear,
    COUNT(CustomerId) AS CntOfCustomer,
    ROUND((COUNT(CustomerId)*100 / (SELECT COUNT(CustomerId) FROM cust_bank_joining)), 2) AS PrcntOfCustomer
FROM cust_bank_joining
GROUP BY JoiningYear
ORDER BY JoiningYear;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

JoiningYear	CntOfCustomer	PrcntOfCustomer
2016	1951	19.51
2017	2143	21.43
2018	2593	25.93
2019	3313	33.13

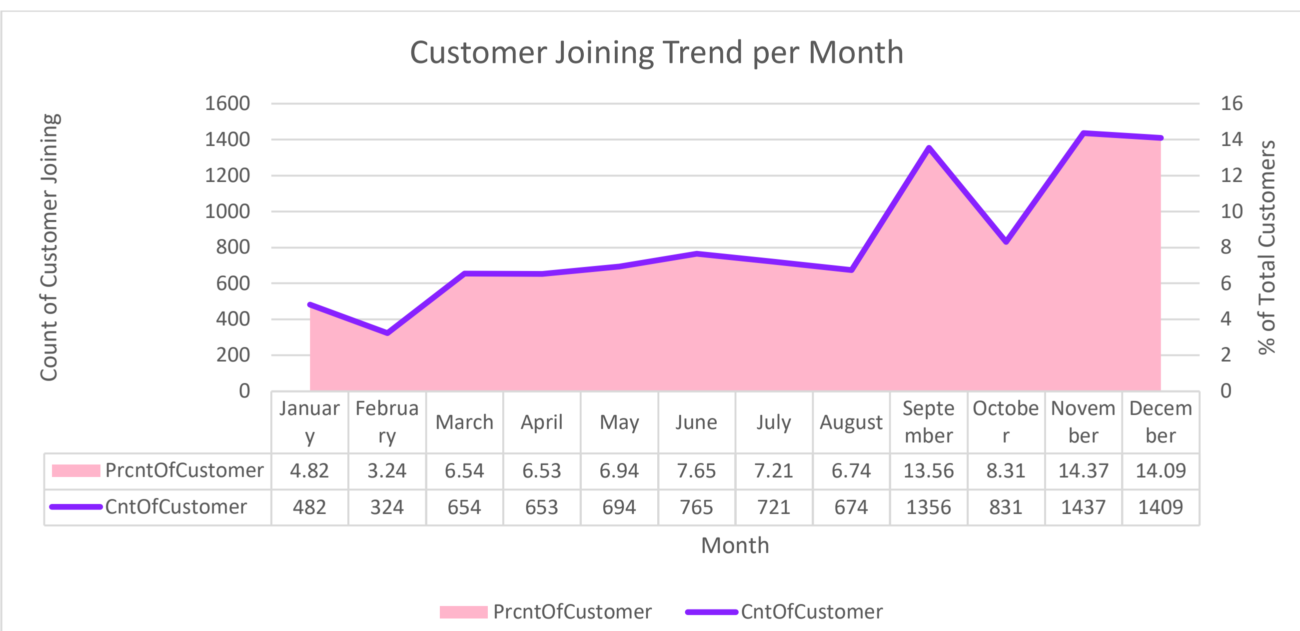


MONTHLY

```sql
-- Monthly
SELECT
    Months AS JoiningMonth,
    COUNT(CustomerId) AS CntOfCustomer,
    ROUND((COUNT(CustomerId)*100 / (SELECT COUNT(CustomerId) FROM cust_bank_joining)), 2) AS PrcntOfCustomer
FROM cust_bank_joining
GROUP BY JoiningMonth
ORDER BY JoiningMonth;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

JoiningMonth	CntOfCustomer	PrcntOfCustomer
1	482	4.82
2	324	3.24
3	654	6.54
4	653	6.53
5	694	6.94
6	765	7.65
7	721	7.21
8	674	6.74
9	1356	13.56
10	831	8.31
11	1437	14.37
12	1409	14.09

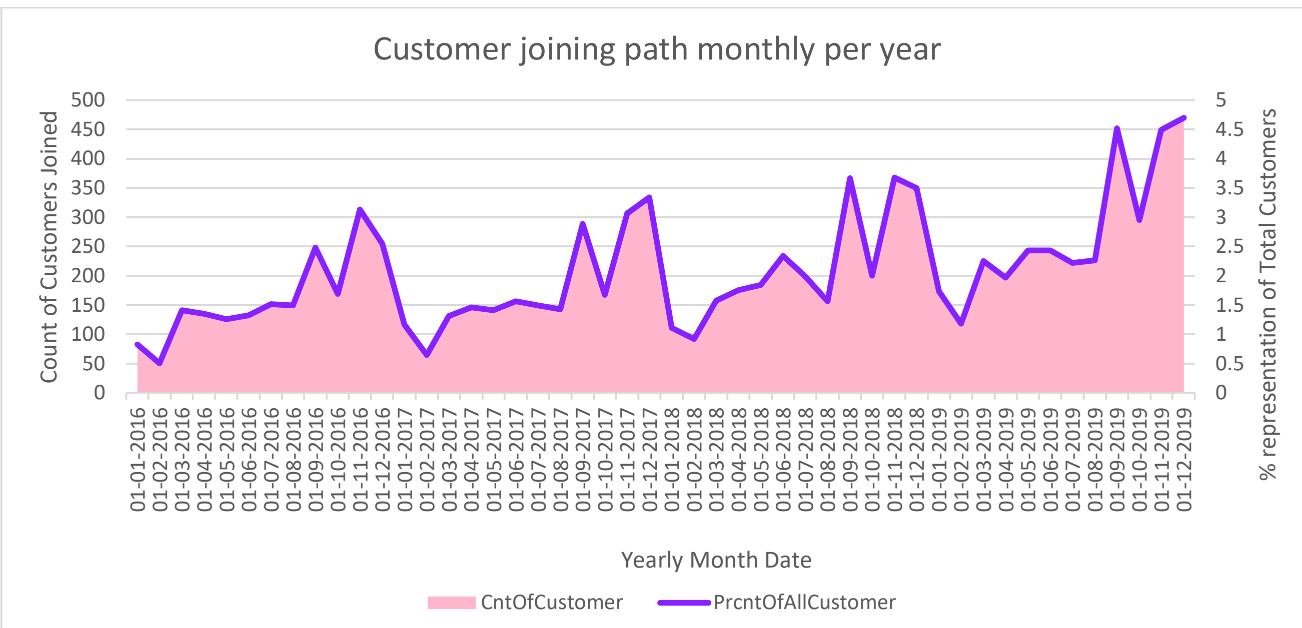


YEARLY MONTH-WISE

```
-- Customer Joining Path
SELECT
    MonthInYears AS Joining,
    COUNT(CustomerId) AS CntOfCustomer,
    ROUND((COUNT(CustomerId)*100 / (SELECT COUNT(CustomerId) FROM cust_bank_joining)), 2) AS PrcntOfAllCustomer
FROM cust_bank_joining
GROUP BY Joining
ORDER BY CAST(RIGHT(Joining, 4) AS SIGNED), CAST(LEFT(Joining, 2) AS SIGNED);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Joining	CntOfCustomer	PrcntOfAllCustomer
01-2016	82	0.82
02-2016	50	0.50
03-2016	141	1.41
04-2016	135	1.35
05-2016	126	1.26
06-2016	132	1.32
07-2016	151	1.51
08-2016	149	1.49
09-2016	248	2.48
10-2016	169	1.69
11-2016	313	3.13
12-2016	255	2.55
01-2017	116	1.16
02-2017	64	0.64
03-2017	131	1.31
04-2017	146	1.46



12. Analyze the relationship between the number of products and the account balance for customers who have exited.

```
SELECT
    NumOfProducts,
    ROUND(AVG(Balance), 2) AS AvgBalance
FROM bank_cust_details
WHERE Exited = 1
GROUP BY NumOfProducts
ORDER BY NumOfProducts;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

NumOfProducts	AvgBalance
1	92028.82
2	90252.36
3	85853.09
4	93733.14

13. Identify any potential outliers in terms of balance among customers who have remained with the bank.

```sql
WITH ranked_balances AS (
    SELECT
        CustomerId,
        Balance,
        PERCENT_RANK() OVER (ORDER BY Balance) AS pr
    FROM bank_cust_details
    WHERE Exited = 0
),

-- Step 2: Get Q1 and Q3 from percent ranks
percentiles AS (
    SELECT
        MAX(CASE WHEN pr <= 0.25 THEN Balance END) AS Q1,
        MAX(CASE WHEN pr <= 0.75 THEN Balance END) AS Q3
    FROM ranked_balances
),

-- Step 3: Compute IQR bounds
bounds AS (
    SELECT
        Q1,
        Q3,
        (Q3 - Q1) AS IQR,
        Q1 - 1.5 * (Q3 - Q1) AS lower_bound,
        Q3 + 1.5 * (Q3 - Q1) AS upper_bound
    FROM percentiles
)

-- Step 4: Identify balances outside IQR bounds (potential outliers)
SELECT
    CustomerId,
    Balance
FROM ranked_balances, bounds b
WHERE Balance < b.lower_bound OR Balance > b.upper_bound;
```

	CustomerId	Balance

no potential outlier in retained customer data using IQR method

14. How many different tables are given in the dataset, out of these tables which table only consists of categorical variables?

There are total 7 tables in the dataset out of which 5 tables consist of only categorical variables. These tables are:

Previous Table Name	Updated Table Name
ActiveCustomer	customer_category
CreditCard	credit_category
ExitCustomer	churn_detail
Gender	gender
Geography	geography

15. **Using SQL, write a query to find out the gender-wise average income of males and females in each geography id. Also, rank the gender according to the average value. (SQL)**

```sql
SELECT
    Location,
    Gender,
    ROUND(AVG(EstimatedSalary), 2) AS AvgIncome,
    RANK() OVER(PARTITION BY Location ORDER BY AVG(EstimatedSalary) DESC) AS Ranking
FROM bank_cust_details
GROUP BY Location, Gender
ORDER BY Location, Ranking;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Location	Gender	AvgIncome	Ranking
France	Male	100174.25	1
France	Female	99564.25	2
Germany	Female	102446.42	1
Germany	Male	99905.03	2
Spain	Female	100734.11	1
Spain	Male	98425.69	2

16. **Using SQL, write a query to find out the average tenure of the people who have exited in each age bracket (18-30, 30-50, 50+).**

```sql
SELECT
    (
        CASE
        WHEN Age BETWEEN 18 AND 30 THEN "18-30"
        WHEN Age BETWEEN 31 AND 50 THEN "31-50"
        WHEN Age > 50 THEN "50+"
        END
    ) AS AgeBracket,
    ROUND(AVG(Tenure)) AS AvgTenure
FROM bank_cust_details
WHERE Exited = 1
GROUP BY AgeBracket;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

AgeBracket	AvgTenure
31-50	5
18-30	5
50+	5

17. Is there any direct correlation between salary and the balance of the customers? And is it different for people who have exited or not?

No, there exists no direct relation between Estimated Salary and Balance of customers since correlation coefficient of these two are at negligible scale

R2 =CORREL(BankCustomerInfo[EstimatedSalary], BankCustomerInfo[Balance])

0.012797 correlation between balance and estimated salary

CustomerId	Surname	EstimatedSalary	CreditScore	Balance
15634602	Hargrave	101348.88	619	0
15647311	Hill	112542.58	608	83807.86
15619304	Onio	113931.57	502	159660.8
15701354	Boni	93826.63	699	0
15737888	Mitchell	79084.1	850	125510.82
15574012	Chu	149756.71	645	113755.78
15592531	Bartlett	10062.8	822	0
15656148	Obinna	119346.88	376	115046.74
15792365	He	74940.5	501	142051.07
15592389	H?	71725.73	684	134603.88
15767821	Bearce	80181.12	528	102016.72
15737173	Andrews	76390.01	497	0
15632264	Kay	26260.98	476	0
15691483	Chin	190857.79	549	0
15600882	Scott	65951.65	635	0
15643966	Goforth	64327.26	616	143129.41
15737452	Romeo	5097.67	653	132602.88
15788218	Henderson	14406.41	549	0

In case of an existing customer, the value is still negligible, but negative sign indicates an inverse relationship between the two variables

Q2 =CORREL(ExitCustomerInfo_1[Balance], ExitCustomerInfo_1[EstimatedSalary])

-0.01254 correlation between salary and balance of exited customer

CustomerId	Surname	EstimatedSalary	Balance	Exited
15634602	Hargrave	101348.88	0	1
15619304	Onio	113931.57	159660.8	1
15574012	Chu	149756.71	113755.78	1
15656148	Obinna	119346.88	115046.74	1
15737452	Romeo	5097.67	132602.88	1
15699309	Gerasimov	118913.53	0	1
15589475	Azikiwe	140469.38	0	1
15794171	Lombardo	27822.99	134264.04	1
15738148	Clarke	181297.65	122522.32	1
15755196	Lavine	194365.76	131394.56	1
15602280	Martin	119708.21	112045.67	1
15771573	Okagbue	117622.8	137843.8	1

18. Is there any correlation between the salary and the Credit score of customers?

No, the correlation coefficient is negligible, but it shows the existence of a negative relationship

R2 =CORREL(BankCustomerInfo[EstimatedSalary], BankCustomerInfo[CreditScore])

-0.00138 correlation between creditscore and estimated salary

CustomerId	Surname	EstimatedSalary	CreditScore	Balance
15634602	Hargrave	101348.88	619	0
15647311	Hill	112542.58	608	83807.86
15619304	Onio	113931.57	502	159660.8
15701354	Boni	93826.63	699	0
15737888	Mitchell	79084.1	850	125510.82
15574012	Chu	149756.71	645	113755.78
15592531	Bartlett	10062.8	822	0
15656148	Obinna	119346.88	376	115046.74
15792365	He	74940.5	501	142051.07
15592389	H?	71725.73	684	134603.88
15767821	Bearce	80181.12	528	102016.72
15737173	Andrews	76390.01	497	0
15632264	Kay	26260.98	476	0

19. **Rank each bucket of credit score as per the number of customers who have churned the bank.**

```sql
WITH cust_credit AS (
    SELECT
        (
            CASE
            WHEN IF(CreditScore<300, 300, CreditScore) BETWEEN 300 AND 579 THEN "300-579 : Poor"
            WHEN CreditScore BETWEEN 580 AND 669 THEN "580-669 : Fair"
            WHEN CreditScore BETWEEN 670 AND 739 THEN "670-739 : Good"
            WHEN CreditScore BETWEEN 740 AND 799 THEN "740-799 : Very Good"
            WHEN IF(CreditScore>850, 850, CreditScore) BETWEEN 800 AND 850 THEN "800-850 : Exceptional"
            END
        ) AS CustomerSegment,
        COUNT(CASE WHEN Exited=1 THEN CustomerId END) AS CntOfChurnedCustomer,
        ROUND((COUNT(CASE WHEN Exited=1 THEN CustomerId END)*100 / (SELECT COUNT(CustomerId) FROM bank_cust_details)), 2) AS PrcntOfChurn
    FROM bank_cust_details
    GROUP BY CustomerSegment
    ORDER BY CustomerSegment
)

SELECT *,
    RANK() OVER(ORDER BY CntOfChurnedCustomer DESC) AS Ranking
FROM cust_credit
ORDER BY Ranking;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerSegment	CntOfChurnedCustomer	PrcntOfChurn	Ranking
580-669 : Fair	685	6.85	1
300-579 : Poor	520	5.20	2
670-739 : Good	452	4.52	3
740-799 : Very Good	252	2.52	4
800-850 : Exceptional	128	1.28	5

20. **According to the age buckets find the number of customers who have a credit card. Also retrieve those buckets that have lesser than average number of credit cards per bucket.**

```sql
SELECT
    (
        CASE
        WHEN Age BETWEEN 18 AND 30 THEN "18-30"
        WHEN Age BETWEEN 31 AND 50 THEN "31-50"
        WHEN Age > 50 THEN "50+"
        END
    ) AS AgeBracket,
    COUNT(CustomerId) AS CntOfCustWithCreditCard
FROM bank_cust_details
WHERE HasCrCard = 1
GROUP BY AgeBracket;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

AgeBracket	CntOfCustWithCreditCard
31-50	4781
18-30	1400
50+	874

LESSER THAN AVERAGE CREDIT CARD AGE BRACKET

```sql
-- lesser than average
WITH credit_card AS (
    SELECT
        (
            CASE
            WHEN Age BETWEEN 18 AND 30 THEN "18-30"
            WHEN Age BETWEEN 31 AND 50 THEN "31-50"
            WHEN Age > 50 THEN "50+"
            END
        ) AS AgeBracket,
        COUNT(CustomerId) AS CntOfCustWithCreditCard
    FROM bank_cust_details
    WHERE HasCrCard = 1
    GROUP BY AgeBracket
)

SELECT *
FROM credit_card
WHERE CntOfCustWithCreditCard < (SELECT ROUND(AVG(CntOfCustWithCreditCard)) FROM credit_card);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

AgeBracket	CntOfCustWithCreditCard
18-30	1400
50+	874

21. Rank the Locations as per the number of people who have churned the bank and average balance of the customers.

```sql
SELECT
    Location,
    COUNT(CustomerId) AS CntOfChurnedCust,
    ROUND(AVG(Balance), 2) AS AvgCustBalance,
    RANK() OVER(ORDER BY COUNT(CustomerId) DESC) AS RankByChurn,
    RANK() OVER(ORDER BY ROUND(AVG(Balance), 2) DESC) AS RankByBalance
FROM bank_cust_details
WHERE Exited = 1
GROUP BY Location
ORDER BY RankByChurn, RankByBalance;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Location	CntOfChurnedCust	AvgCustBalance	RankByChurn	RankByBalance
Germany	814	120361.08	1	1
France	810	71192.8	2	3
Spain	413	72513.35	3	2

22. **As we can see that the “CustomerInfo” table has the CustomerID and Surname, now if we have to join it with a table where the primary key is also a combination of CustomerID and Surname, come up with a column where the format is “CustomerID_Surname”.**

```sql
SELECT CONCAT(CustomerId, "_", Surname) AS CombinedKey
FROM bank_cust_details;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fe

CombinedKey
15565879_Riley
15566139_Ts'ui
15566251_Ferrari
15566295_Sanders
15566563_Duigan

23. **Without using “Join”, can we get the “ExitCategory” from ExitCustomers table to Bank_Churn table? If yes do this using SQL.**

```sql
SELECT *,
    (
        SELECT ExitCategory FROM churn_detail c
        WHERE b.Exited = c.ExitId
    ) AS ExitCategory
FROM bank_churn b;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

CustomerId	CreditScore	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	Exited	ExitCategory
15565701	698	4	161993.89	1	0	0	0	Retain
15565706	612	4	0	1	1	1	1	Exit
15565714	601	3	64430.06	2	0	1	0	Retain
15565779	627	4	57809.32	1	1	0	0	Retain
15565796	745	4	96048.55	1	1	0	0	Retain

24. **Were there any missing values in the data Using which tool did you replace them, and what are the ways to handle them?**

There are no missing values in the data. This was verified in Excel by using the filter option, which allows us to filter columns based on conditions, and we found no 'blanks' option in that filter section.

However, when loading data into Power BI, some null columns were imported. Therefore, in the Power Query editor, we used the “Keep Top Row” feature to extract only the required rows.

25. Write the query to get the customer IDs, their last name, and whether they are active or not for the customers whose surname ends with "on".

```sql
SELECT
    b.CustomerId,
    b.Surname AS LastName,
    c.ActiveCategory AS ActiveOrNot
FROM bank_cust_details b
JOIN customer_category c
    ON b.IsActiveMember = c.ActiveId
WHERE Surname LIKE "%on"
ORDER BY CustomerId;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerId	LastName	ActiveOrNot
15566495	Hanson	Inactive Member
15567839	Gordon	Active Member
15568088	Jamieson	Inactive Member
15568360	Rolon	Active Member
15570289	Benson	Inactive Member
15571193	Morrison	Active Member
15571305	Stephenson	Inactive Member
15572145	Ashton	Inactive Member
15572415	Preston	Active Member
15573599	Adamson	Active Member
15573730	Thompson	Inactive Member

26. Can you observe any data disrupency in the Customer's data? As a hint it's present in the IsActiveMember and Exited columns. One more point to consider is that the data in the Exited Column is absolutely correct and accurate.

Yes, there is a possibility of a discrepancy in the data. As we can see, some customers who have exited are still marked as Active Members, which can lead our analysis in the wrong direction.

This also raises a concern whether the customers have exited or not because many of these customers still have a good amount of balance in their account.

```sql
SELECT *
FROM bank_churn
WHERE Exited = 1 AND IsActiveMember = 1;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

CustomerId	CreditScore	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	Exited
15565706	612	4	0	1	1	1	1
15566312	660	3	0	3	1	1	1
15566378	515	6	129387.94	1	0	1	1
15566531	724	4	88046.88	1	0	1	1
15567063	766	3	106434.94	1	0	1	1
15567506	738	3	114940.67	2	1	1	1
15567778	690	3	144027.8	1	1	1	1
15567980	537	4	100727.5	1	0	1	1
15568573	554	5	105701.91	1	0	1	1
15568748	671	4	99564.22	1	1	1	1
15568819	619	3	132796.04	3	1	1	1

SUBJECTIVE QUESTIONS

1. Customer Behaviour Analysis: What patterns can be observed in the spending habits of long-term customers compared to new customers, and what might these patterns suggest about customer loyalty?

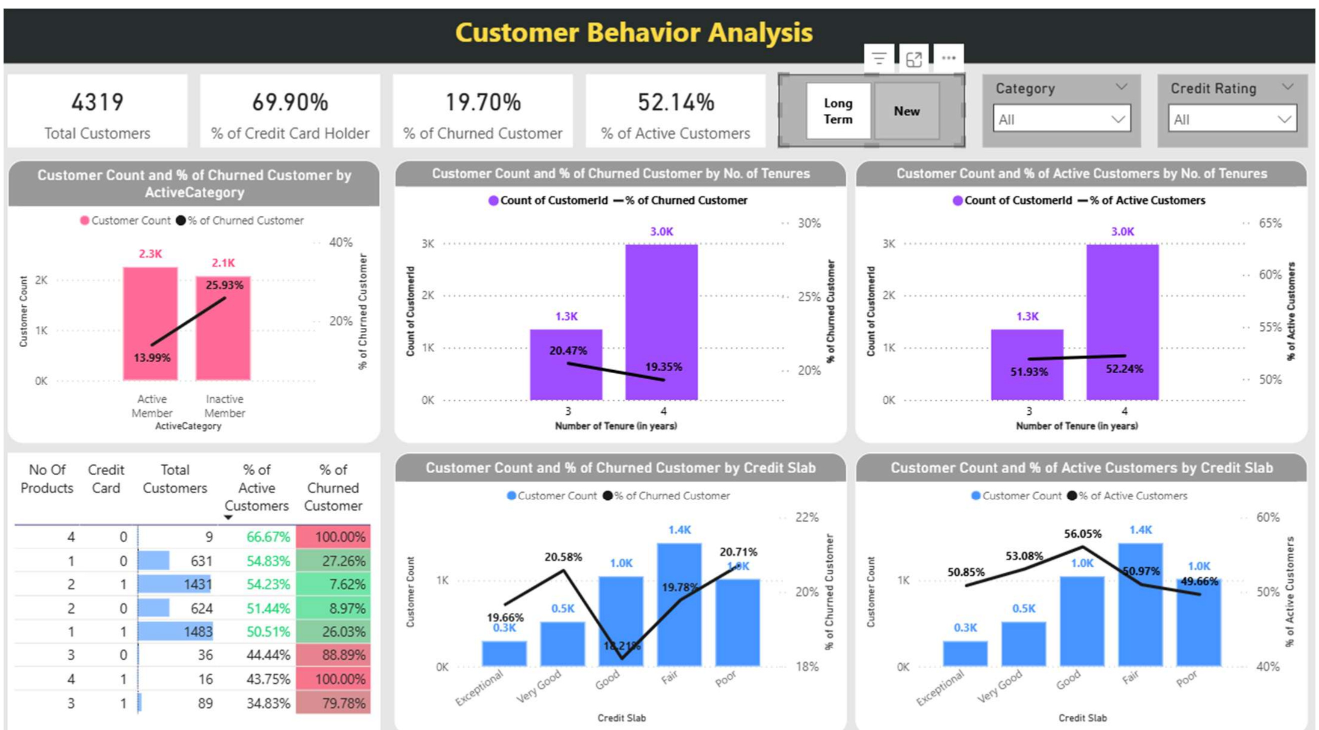
APPROACH

- Assuming customers who have been with the bank for more than 4 years are long-term, and the rest are new
- Finding the count of customers in different tenure periods, the number of products held, the activeness, and the credit slab
- Comparing groups based on % of churned customers, % of active members and % of credit card holders
- At last, providing slicers of long-term and new customers to analyse their behaviour
- Also adding a card visual to find the overall churn rate amongst the group

overall dashboard

long-term customer-specific

new customer-specific



INSIGHTS

- There are 10000 customers, out of which 51.51% customers are our active customers, 71.04% customers have a credit card, and the churn rate stands at 20.88%
- Out of 10000 customers, 56.8% are long-term customers, and 43.2% customers are new customers
- For long-term customers, they tend to have more credit card holders, fewer active customers, and a higher churn rate compared to new customers.
  Most long-term customers typically have either one or two products. For those with more than two products, the churn rate increases as the number of products goes up, active users decrease with more products, and the number of credit card holders remains nearly unchanged.
- For new customers, credit card holders tend to be less numerous, more active, and have a lower churn rate compared to long-term customers. Most new customers have either one or two products. As the number of products increases, the churn rate also increases, but active customers are more common with four products, while credit card holders decrease.

RECOMMENDATIONS

- ✓ The above insights suggest that a larger number of products leads to a higher churn rate and less customer activity.
- ✓ Long-term customers have less activity, especially those having a higher no. of products. Long-term customers with a credit card are more dominant.
- ✓ New customers with credit cards are fewer and highly active when they have 4 products (important characteristics), but the credit card share is low amongst their products.

2. **Product Affinity Study: Which bank products or services are most commonly used together, and how might this influence cross-selling strategies?**

APPROACH

- We will take the number of products held by customers and combine it with the fact of whether they have a credit card or not
- Then analysing churn rates in different bundles and the count of customers in each bundle

Product Affinity

```sql
SELECT
  NumOfProducts,
  HasCrCard,
  COUNT(CustomerId) AS cnt,
  ROUND(
    SUM(
      CASE WHEN Exited = 1 THEN 1 ELSE 0 END
    )*100 / COUNT(CustomerId)
  , 2) AS percent_of_churn
FROM bank_cust_details
GROUP BY NumOfProducts, HasCrCard
ORDER BY NumOfProducts, HasCrCard;
```

NumOfProducts	HasCrCard	cnt	percent_of_churn
1	0	1506	27.76
1	1	3578	27.70
2	0	1344	8.33
2	1	3246	7.27
3	0	76	84.21
3	1	190	82.11
4	0	19	100.00
4	1	41	100.00

OR using a table visual in POWER BI combined with DAX

No Of Products	Credit Card	Tot Customers	Tot Churned Customers	% of Churned Customer
2	1	3246	236	7.27%
2	0	1344	112	8.33%
1	1	3578	991	27.70%
1	0	1506	418	27.76%
3	1	190	156	82.11%
3	0	76	64	84.21%
4	0	19	19	100.00%
4	1	41	41	100.00%

INSIGHTS

- Customers who have four products have a 100% churn rate, indicating that they are risky customers and we should not cross-sell in this category
- Customers who have 3 or 1 product still have a higher churn rate if they have a credit card as a product
- Customer has 2 products, out of which one is a credit card, which has the least churn rate, indicating a strong cross-selling opportunity

RECOMMENDATIONS

- ✓ We can cross-sell to customers who have only 1 product (do not have a credit card) to opt for a credit card, and if they have one, then give them additional benefits or an upgraded card
- ✓ We can encourage customers who have 2 products and do not use a credit card to use one and offer them higher reward cards

3. **Geographic Market Trends: How do economic indicators in different geographic regions correlate with the number of active accounts and customer churn rates?**

APPROACH

- Using a matrix visual to understand what different indicators tell about different geographical regions
- Using DAX formulas to find out % of active customer and % of churned customers
- Finally adding some conditional formatting to capture insights faster

GeographyLocation	Tot Customers	% of Credit Card Holder	Avg Income	Avg Credit Score	% of Active Customers	% of Churned Customer
Spain	2477	69.48%	99,440.57	651.33	52.97%	16.67%
France	5014	70.66%	99,899.18	649.67	51.68%	16.15%
Germany	2509	71.38%	1,01,113.44	651.45	49.74%	32.44%

INSIGHTS

- France constitutes approximately. HALF of the total customers followed by Spain and Germany, contributing ONE-FOURTH of the total customers
- Germany has the highest percentage of credit card holders, i.e., 71.38% and the highest Average Income, i.e., 1L, amongst all three geographical locations
- Average Credit Score of all Regions is approximately the same in the range 648 to 652, i.e., under the fair category
- Germany has the highest churn rate, i.e., 32.44% with the lowest active customer approx. 49.74%

RECOMMENDATIONS

✓ If a customer belongs to Germany, they have a high probability of getting churned about 0.3 percent chances of getting churned

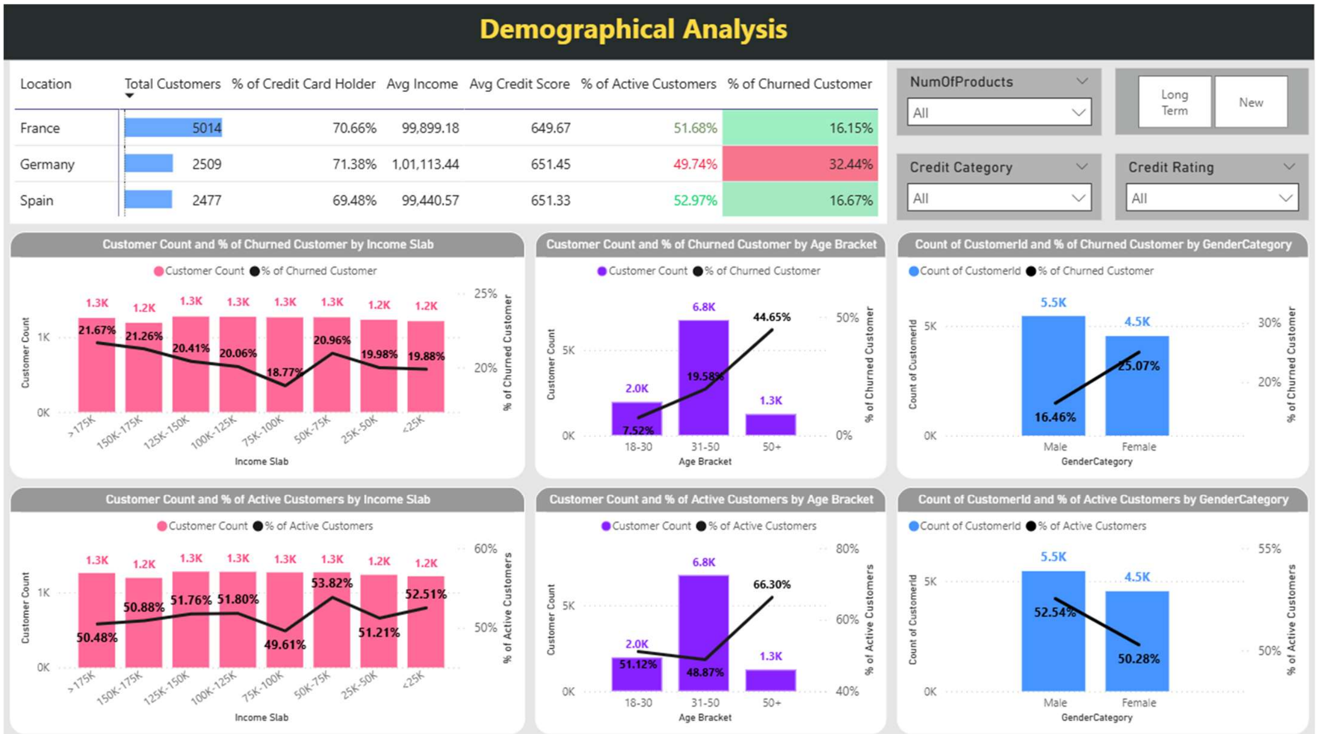
4. **Risk Management Assessment: Based on customer profiles, which demographic segments appear to pose the highest financial risk to the bank, and why?**

APPROACH

- Using DAX to first segregate customers in different segments based on tenure, age, income, and credit score
- Using visualizations to find the segment that has the highest chances of getting churn
- Combining it with geographical analysis to get a better understanding of the churned and inactive segments

```
customer_segment =
    VAR Div1 = 25000
  VAR Div2 = 50000
    VAR Div3 = 75000
    VAR Div4 = 100000
    VAR Div5 = 125000
    VAR Div6 = 150000
    VAR Div7 = 175000
RETURN
  ADDCOLUMNS(
    SELECTCOLUMNS(customer_detail, "CustomerId", customer_detail[CustomerId]),
    "Customer Bracket", IF(
      LOOKUPVALUE(bank_churn[Tenure], bank_churn[CustomerId], [CustomerId]) > 4,
      "Long Term",
      "New"
    ),
    "Age Bracket", VAR CustAge = LOOKUPVALUE(customer_detail[Age],
customer_detail[CustomerId], [CustomerId])
      RETURN SWITCH(
        TRUE(),
        CustAge > 50, "50+",
        CustAge > 30, "31-50",
        "18-30"
      ),
    "Income Bracket", VAR CustSalary =
LOOKUPVALUE(customer_detail[EstimatedSalary],customer_detail[CustomerId], [CustomerId])
      RETURN SWITCH(
        TRUE(),
        CustSalary <= Div1, "<25K",
        CustSalary <= Div2, "25K-50K",
                CustSalary <= Div3, "50K-75K",
                CustSalary <= Div4, "75K-100K",
                CustSalary <= Div5, "100K-125K",
                CustSalary <= Div6, "125K-150K",
                CustSalary <= Div7, "150K-175K",
                ">175K"
      ),
    "Credit Bracket", VAR CustCredit = LOOKUPVALUE(bank_churn[CreditScore],
bank_churn[CustomerId], [CustomerId])
      RETURN SWITCH(
        TRUE(),
        CustCredit >= 800, "Exceptional",
        CustCredit >= 740, "Very Good",
        CustCredit >= 670, "Good",
        CustCredit >= 580, "Fair",
        "Poor"
      )
)
```

dashboard



INSIGHTS

- Higher income groups have a higher churn rate and low active customers, i.e., income above 100K, but there is a specific income group that has high active customers with a high churn rate, which is the 50K-75K income group
- According to age segmentation, people with 50+ have a high active customer and high churn rate (which is opposite to the general trend of high churn, low activity norm), while the age group 18-30 has the lowest churn rate and
- Credit Ratings depict the natural relationships between churn and active members, i.e., churn rate is high when active rate is low and vice versa. And churn rate increases as we move from the good to the poor category of credit score

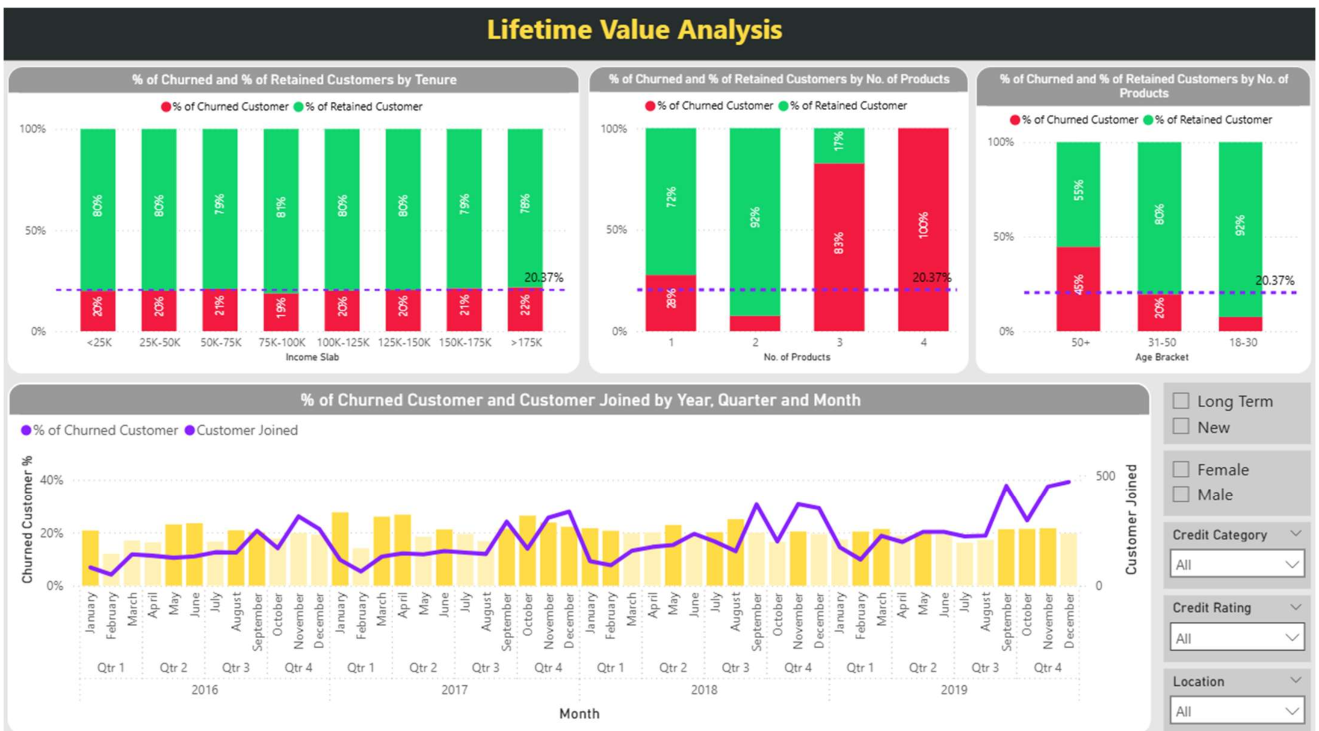
RECOMMENDATION

✓ High Income groups with credit ratings below and equal to Good, having 50+ age, are the group of customers that have the highest chances of churn
✓ Income slab 50K-75K and age group 50+ are abnormal cases of churn in which active customers are high, and even churn is high
✓ Female customers have a higher churn rate than male customers

5. **Customer Tenure Value Forecast: How would you use the available data to model and predict the lifetime (tenure) value in the bank of different customer segments?**

APPROACH

- First analysing the trend of customer joining and churned customer on monthly, quarterly and yearly basis. Using line chart for joining trend and bar chart for churn additionally highlighting months that have churn rate higher than average churn rate of 20.37%
- Second analysing % churned customer by income, no. of products and age group and comparing which segments have a higher churn rate than baseline value of 20.37%
- Third providing additional filter options to compare trends between different customer segments.



INSIGHTS

- The overall joining trend is a rising graph indicating that we are onboarding people. Churn rate was higher in 2017 first and last quarter as compared to overall trend
- There is an abrupt increase in customer who joined during Qtr3 and Qtr4 of every year. While Qtr1 and Qtr2 have a steady increase.
- We lost most customers during time period between Qtr4 2017 - Qtr1 2018
- Income Slab 150K-175K and >175K have a higher churn rate than the baseline average churn rate

- Customers with 4 products have 100% churn rate, followed by 3 products. While customers with 1 product are near to baseline average churn and 2 product customers have the least churn and most successful customers
- About HALF of the customers in the age group 50+ have churned. While churn rate is below the baseline average for 31-50 and quite low for 18-30

RECOMMENDATIONS

✓ Following is the persona of customers who can churn and are risky customers who is a female aged above 50+, having 4 products or 3 products, situated in Germany, having income 125K and above and recently inactive

6. **Marketing Campaign Effectiveness: How could you assess the impact of marketing campaigns on customer retention and acquisition within the dataset? What extra information would you need to solve this?**

We can evaluate the effectiveness of marketing by creating a trend line chart that shows the number of customers who joined on a monthly, quarterly, and yearly basis.

This can be implemented using a calendar table that establishes a relationship with the BankDOJ column. Additionally, we can use a Column Line chart where the line represents customers who joined through the BankDOJ column, and the columns show the percentage of customers churned, using a simple measure. Moreover, conditional formatting can be applied to highlight months or years in which the churn rate exceeds the average churn rate of 20.37%.

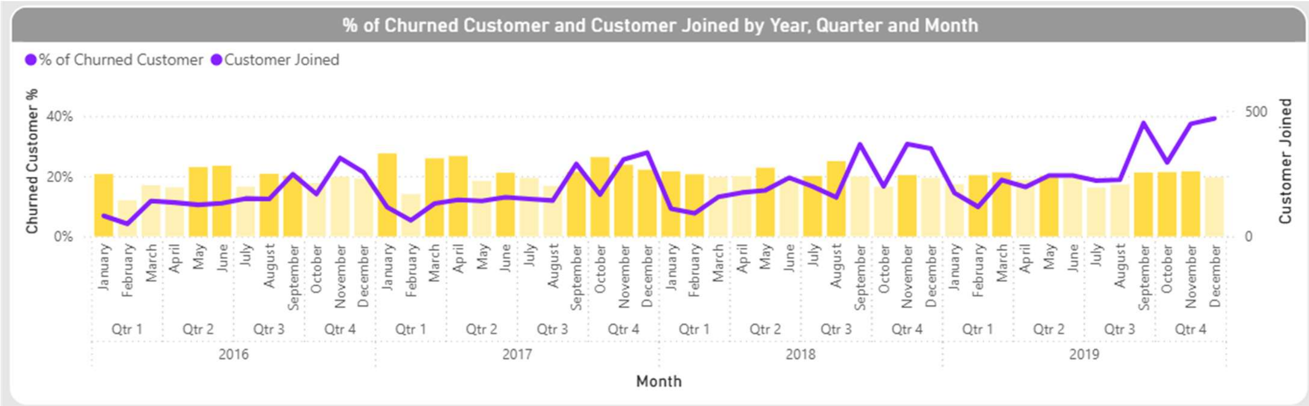
calendar DAX

```
Calendar =
  VAR MinDate = MIN(customer_detail[Bank DOJ])
  VAR MaxDate = MAX(customer_detail[Bank DOJ])
RETURN
  ADDCOLUMNS(
    CALENDAR(MinDate, MaxDate),
    "Year", YEAR([Date]),
    "Quarter", CONCATENATE("Q", QUARTER([Date])),
    "Week of Year", WEEKNUM([Date]),
    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "mmm"),
    "Day of Week Number", WEEKDAY([Date]),
    "Day of Week Name", FORMAT([Date], "ddd"),
    "Day of Month", DAY([Date])
  )
```

measure DAX to find % of churned customers

```
% of Churned Customer = DIVIDE(
        CALCULATE(COUNT(customer_detail[CustomerId]), bank_churn[Exited] = 1),
        COUNT(customer_detail[CustomerId])
    )
```

graphical output



In addition to all these, we would require marketing data such as duration of campaign, schedule of campaign, schedule of campaign, etc. to clarify and assess if marketing did well or not.

The line chart will guide us about the acquisition of customers, and the column chart (churn rate) will express the retention rate of customers

7. **Customer Exit Reasons Exploration: Can you identify common characteristics or trends among customers who have exited that could explain their reasons for leaving?**

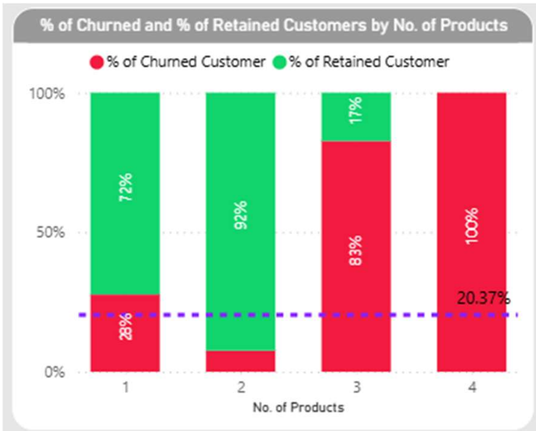
The following are the major factors that are responsible for the exit of customers, as per their influence on churn rate, ordered from highly influential to least influential:

1) **Number of Products:**
   As customers indulge in more and more products, churn rate increases drastically, and 100% churn in customers having 4 products, with the exception that if a customer has 2 products, they have the least churn i.e. 7%
2) **Age Group:**
   The older a person is higher their chances of being churned with customers who have 50+, having a churn rate of 45%
3) **Location:**
   If a person belongs to Germany, then we can say 1 out of every 3 customers has a high probability of getting churned
4) **Gender:**
   Female customers have a higher churn rate than male customers, approximately 10% higher

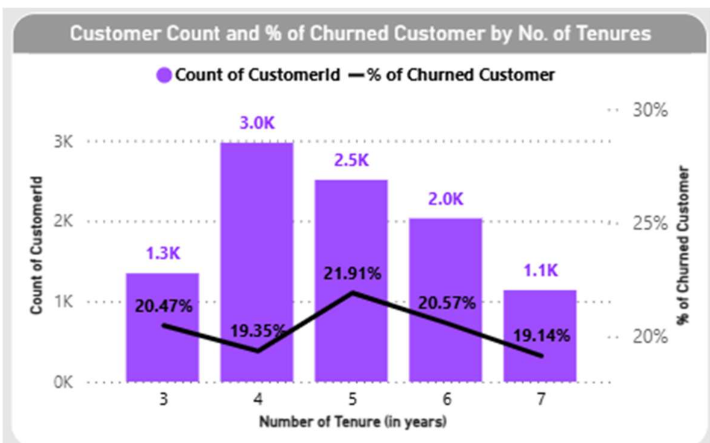
8. **Are 'Tenure', 'NumOfProducts', 'IsActiveMember', and 'EstimatedSalary' important for predicting if a customer will leave the bank?**

Yes, all of them can become good indicators to predict customer churn. This can be graphically validated through the following charts:

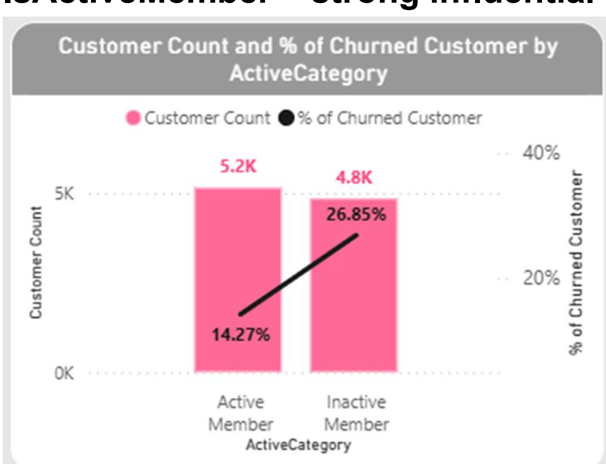
- **Number of Products – highly influential factor**



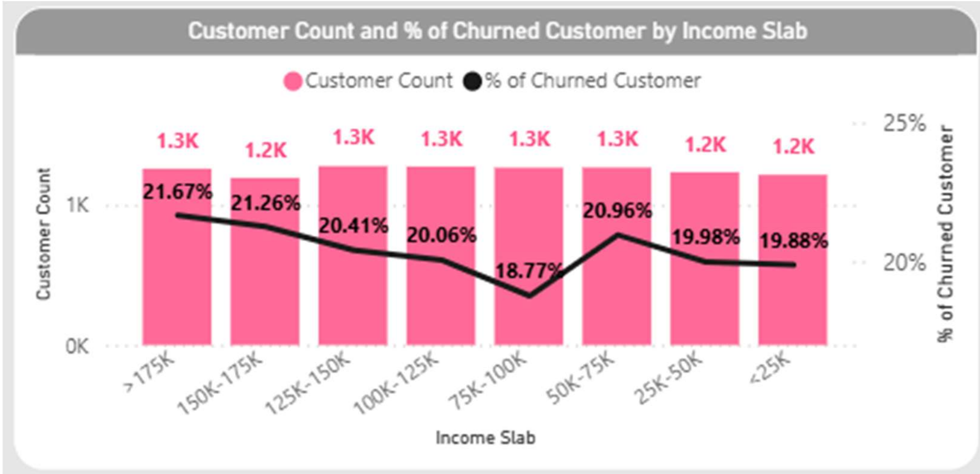
- **Tenure – a considerable factor**



- **IsActiveMember – strong influential factor**



- **Estimated Salary – a considerable factor**



9. Utilize SQL queries to segment customers based on demographics and account details.

```sql
SELECT
  bc.CustomerId,
    bc.Surname,
    bc.Gender,
    bc.Location,
    bc.BankDOJ,
    bc.Tenure,
  bc.NumOfProducts,
  (
    CASE
        WHEN bc.EstimatedSalary < 25000 THEN "<25K"
        WHEN bc.EstimatedSalary <= 50000 THEN "25K-50K"
        WHEN bc.EstimatedSalary <= 75000 THEN "50K-75K"
        WHEN bc.EstimatedSalary <= 100000 THEN "75K-100K"
        WHEN bc.EstimatedSalary <= 125000 THEN "100K-125K"
        WHEN bc.EstimatedSalary <= 150000 THEN "125K-150K"
        WHEN bc.EstimatedSalary <= 175000 THEN "150K-175K"
        ELSE ">175K"
        END
    ) AS IncomeSlab,
    (
    CASE
        WHEN Age <= 30 THEN "18-30"
        WHEN Age <= 50 THEN "31-50"
        ELSE "50+"
        END
    ) AS AgeBracket,
  (
    CASE
    WHEN IF(CreditScore<300, 300, CreditScore) BETWEEN 300 AND 579 THEN "300-579 : Poor"
    WHEN CreditScore BETWEEN 580 AND 669 THEN "580-669 : Fair"
    WHEN CreditScore BETWEEN 670 AND 739 THEN "670-739 : Good"
    WHEN CreditScore BETWEEN 740 AND 799 THEN "740-799 : Very Good"
    WHEN IF(CreditScore>850, 850, CreditScore) BETWEEN 800 AND 850 THEN "800-850 : Exceptional"
    END
  ) AS CustomerSegment,
    crc.Category AS Is_CreditCard,
  cc.ActiveCategory AS Is_Active,
    cd.ExitCategory AS Is_Exited
FROM bank_cust_details bc
JOIN churn_detail cd
  ON bc.Exited = cd.ExitId
JOIN customer_category cc
  ON bc.IsActiveMember = cc.ActiveId
JOIN credit_category crc
  ON bc.HasCrCard = crc.CreditId;
```

CustomerId	Surname	Gender	Location	BankDOJ	Tenure	NumOfProducts	IncomeSlab	AgeBracket	CustomerSegment	Is_CreditCard	Is_Active	Is_Exited
15565714	Cattaneo	Male	France	2019-10-02	3	2	75K-100K	31-50	580-669 : Fair	non credit card holder	Active Member	Retain
15566292	Okwuadigbo	Male	Spain	2016-01-10	7	2	50K-75K	31-50	300-579 : Poor	non credit card holder	Active Member	Retain
15566740	Nazarova	Male	Spain	2018-05-10	5	1	125K-150K	50+	580-669 : Fair	non credit card holder	Active Member	Retain
15566790	McIntyre	Male	France	2019-03-25	4	2	25K-50K	18-30	580-669 : Fair	non credit card holder	Active Member	Retain
15567328	Ch'en	Male	Spain	2017-04-07	6	1	<25K	31-50	670-739 : Good	non credit card holder	Active Member	Retain
15567367	Tao	Female	Germany	2018-03-15	5	1	100K-125K	31-50	580-669 : Fair	non credit card holder	Active Member	Retain
15567383	Slone	Female	Germany	2016-07-27	7	2	25K-50K	31-50	670-739 : Good	non credit card holder	Active Member	Retain
15567431	Kodilinyechukwu	Male	France	2016-10-13	6	1	>175K	50+	740-799 : Very Good	non credit card holder	Active Member	Retain
15567839	Gordon	Male	France	2019-11-21	3	1	75K-100K	31-50	300-579 : Poor	non credit card holder	Active Member	Retain
15567919	Lazarev	Male	Germany	2018-09-29	4	2	100K-125K	31-50	580-669 : Fair	non credit card holder	Active Member	Retain

10. **How can we create a conditional formatting setup to visually highlight customers at risk of churn and to evaluate the impact of credit card rewards on customer retention?**

APPROACH

- Making a new visual calculation in our table visual using DAX that will combine factors that contribute to churn of customers
- ▪ Using conditional formatting to add a red circle for all exited customers and then applying another conditional formatting to highlight the customer ID if the “Is Risky” value is greater than 3.

```
Is Risky =
VAR RiskScore = COALESCE(
    SWITCH(
    TRUE(),
    [No. Of Products] = 4, 2,
    [No. Of Products] = 3, 1,
    [No. Of Products] = 2, -1
) + SWITCH(
    TRUE(),
    [Location] = "Germany", 1
) + SWITCH(
    TRUE(),
    [Credit Bracket] = "Poor", 2,
    [Credit Bracket] = "Fair", 1
) + SWITCH(
    TRUE(),
    [Age Bracket] = "50+", 1
) + SWITCH(
    TRUE(),
    [Is Active] = "Inactive Member", 1
),
0
)
RETURN
    RiskScore
```

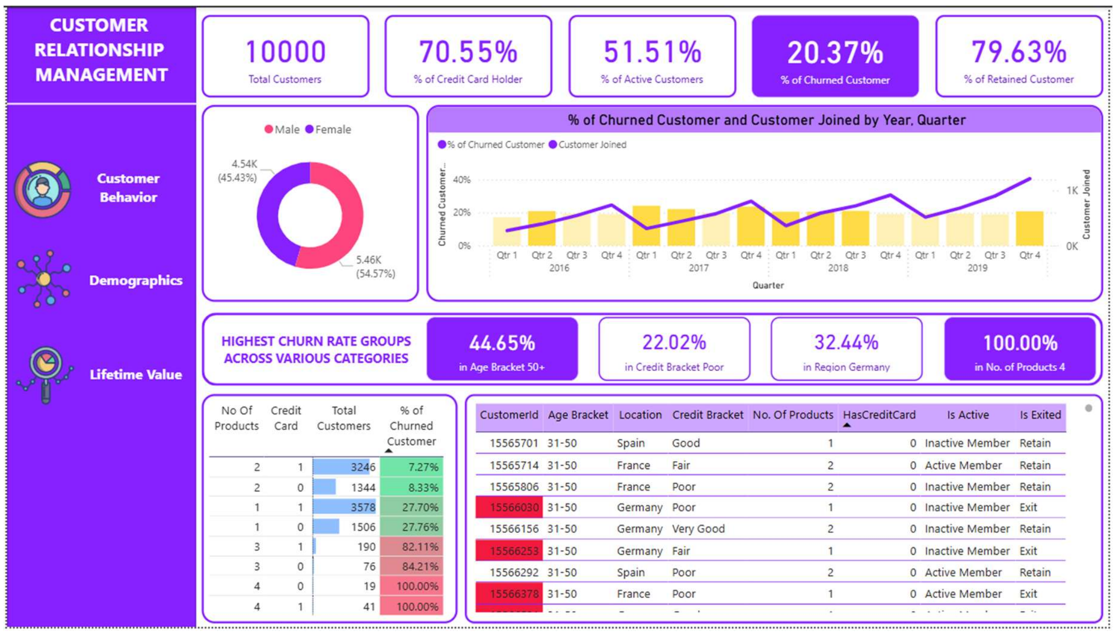
RESULT

CustomerId	Age Bracket	Location	Credit Bracket	No. Of Products	HasCreditCard	Is Active	Is Exited
15595917	31-50	France	Fair	1	0	Active Member	Retain
15595937	31-50	Germany	Poor	2	0	Inactive Member	Retain
15596013	50+	Germany	Good	1	0	Inactive Member	Exit
15596088	31-50	France	Good	2	0	Active Member	Retain
15596212	31-50	Spain	Very Good	2	0	Inactive Member	Retain
15596339	50+	France	Poor	1	0	Active Member	Retain
15596379	31-50	Germany	Very Good	1	0	Active Member	Retain
15596455	31-50	Spain	Poor	1	0	Inactive Member	Exit

11. What is the current churn rate per year and overall in the bank? Can you suggest some insights to the bank about which kind of customers are more likely to churn and what different strategies can be used to decrease the churn rate?

Year	% of Churned Customer	% of Active Customers
2016	19.27%	50.79%
2017	22.35%	51.05%
2018	20.21%	51.83%
2019	19.86%	51.98%
Total	**20.37%**	**51.51%**

12. Create a dashboard incorporating all the KPIs and visualization-related metrics. Use a slicer in order to assist in selection in the dashboard.



13. How would you approach this problem if the objective and subjective questions weren't given?

- First identifying demographics and composition of data to find out who our customers are, where they are located, and what their financial situations are.
- Then, using churn, credit card, ratings, and inactive parameters to find out drop rates and user inactivity segments
- Finally finding similarities of customers joining or usage patterns to get personas of customers who are Risky or on the verge of churn

14. In the "Bank_Churn" table, how can you modify the name of the "HasCrCard" column to "Has_creditcard"?

```sql
ALTER TABLE bank_churn
RENAME COLUMN HasCrCard TO Has_creditcard;
```